Quantitative Finance, 2021
Vol. 21, No. 11, 1809–1824, https://doi.org/10.1080/14697688.2021.1912379

CMS spread options

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(*Received 4 June 2020; accepted 26 March 2021; published online 15 June 2021*)

We analyze European options on CMS spreads, obtaining closed form formulas for the values of these instruments. There are three key steps in this analysis. The first step is to create a hybrid numeraire in which the spread $\tilde{R}_1(T) - \tilde{R}_2(T)$ is a Martingale. Like other CMS calculations, this reduces valuation to the valuation of standard European options plus quadratic convexity corrections. The second step is to combine the volatility smiles of the individual swap rates $\tilde{R}_1$ and $\tilde{R}_2$ to obtain the smile of the spread $\tilde{R}_1(T) - \tilde{R}_2(T)$. We do this by modeling the volatility smiles of both swap rates by normal ($\beta = 0$) SABR models. (In practice this may require using a shifter to convert SABR models with $\beta \neq 0$ into SABR models with $\beta = 0$.) We then show that the volatility smile of the spread $\tilde{R}_1(T) - \tilde{R}_2(T)$ is also governed by a normal SABR model, and derive the SABR parameters for the spread. The third step is to use the closed-form formulae for quadratic options under the SABR model to obtain explicit formulae for the valuation of European options on CMS spreads. These formulas are not exact, but they are accurate up to $O(\varepsilon^2)$, the same accuracy as the original SABR formulas. They also satisfy call-put parity exactly, and are exactly consistent with the valuation of CMS options on the component swap rates $\tilde{R}_1$ and $\tilde{R}_2$.

Keywords: CMS rates; Spread options; SABR model

1. Introduction

We analyze European options on CMS spreads. The payoffs of caplets, floorlets, and swaplets on CMS spreads are

$$V_c^S(t_{ex}, K) = Z(t_{ex}; T_0)\left[\tilde{R}_1(t_{ex}) - \tilde{R}_2(t_{ex}) - K\right]_+, \qquad (1a)$$

$$V_p^S(t_{ex}, K) = Z(t_{ex}; T_0)\left[K - \left(\tilde{R}_1(t_{ex}) - \tilde{R}_2(t_{ex})\right)\right]_+, \qquad (1b)$$

$$V_s^S(t_{ex}, K) = Z(t_{ex}; T_0)\left[\tilde{R}_1(t_{ex}) - \tilde{R}_2(t_{ex}) - K\right], \qquad (1c)$$

where t_{ex} is the option exercise date, K is the strike, and $Z(t_{ex}; T_0)$ is the discount factor for the settlement date T_0 as seen at the expiry date t_{ex}. Here $\tilde{R}_1(t_{ex})$ and $\tilde{R}_2(t_{ex})$ are the swap rates for the intervals T_0 to T_{n_1} and T_0 to T_{n_2}, as seen at t_{ex}. Our objective is to obtain closed form formulae for the values $V_c^S(t,K)$, $V_p^S(t,K)$, $V_s^S(t,K)$ of these European options.

Digital calls (puts) on CMS spreads pay 1 if the spread exceeds (is less than) the strike K at expiry,

$$V_{dc}^S(t_{ex}, K) = Z(t_{ex}; T_0)\cdot 1_{\tilde{R}_1(t_{ex}) - \tilde{R}_2(t_{ex}) > K} = -\frac{\partial}{\partial K}V_c^S(t_{ex}, K), \qquad (2a)$$

$$V_{dp}^S(t_{ex}, K) = Z(t_{ex}; T_0)\cdot 1_{\tilde{R}_1(t_{ex}) - \tilde{R}_2(t_{ex}) < K} = \frac{\partial}{\partial K}V_p^S(t_{ex}, K). \qquad (2b)$$

Consequently, digital options on CMS spreads can be valued for all dates t by differentiating with respect to the strike K, either analytically or numerically,

$$V_{dc}^S(t, K) = -\frac{\partial}{\partial K}V_c^S(t, K), \qquad (3a)$$

$$V_{dp}^S(t, K) = \frac{\partial}{\partial K}V_p^S(t, K). \qquad (3b)$$

Accordingly, we do not need to consider digital options further.

A key motivation for valuing these CMS options arises from Bermudans. A popular Bermudan is a callable swap whose 'coupon leg' is composed of either digital or European options on CMS spreads. One cannot hope to correctly price or hedge these Bermudans unless the term structure model correctly prices the underlying European CMS spread options. This requires either calibrating the term structure model to these spread options, or using an adjuster (Hagan 2002) to match the payoffs to the values of these options. Either

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approach requires the ability to price European options on CMS spreads.

There are three key steps in our analysis. The first step is to create a numeraire in which the spread

$$\tilde{S}(T) = \tilde{R}_1(T) - \tilde{R}_2(T) \tag{4}$$

is a Martingale. Like other CMS calculations, this reduces valuation to the valuation of standard European options on the spread $\tilde{S}(T)$, plus the valuation of quadratic options which arise from the convexity corrections (Hagan 2003, Andersen and Piterbarg 2010, Hagan *et al.* 2020).

The second step is to combine the volatility smiles of the individual swap rates $\tilde{R}_1$ and $\tilde{R}_2$ to obtain the volatility smile of the spread $\tilde{S}(T)$. We do this by modeling the volatility smiles of both swap rates by *normal* ($\beta = 0$) SABR models. Since a desk may be using $\beta \neq 0$ to manage these smiles, this may require using a *shifter* which converts the SABR models with $\beta \neq 0$ to SABR models with $\beta = 0$ (see Appendix 3). We then use singular perturbation techniques to show that the volatility smile of the spread $\tilde{S}(T)$ is also governed by a normal SABR model, at least up to $O(\varepsilon^2)$, and derive the SABR parameters for the spread (Hagan *et al.* forthcoming).

The third step is to use the closed-from formulae for vanilla and quadratic options under the SABR model (Hagan *et al.* 2002, 2016, 2019) to obtain explicit formulae for the valuation of European options on CMS spreads. These formulas are not exact, but they are accurate up to $O(\varepsilon^2)$, the same accuracy as the original SABR formulae. They also satisfy call-put parity exactly.

As we shall see, our approach ensures that the valuation of CMS spreads is exactly consistent with the valuation of CMS options on the component swap rates $\tilde{R}_1$ and $\tilde{R}_2$, regardless of the methodology used. The sole requirement is that these components satisfy call-put parity. This allows CMS options on swap rates and swap rate spreads to be hedged together in a consistent manner.

Currently, the standard approach to pricing CMS spread options uses copulas (Kienitz 2011, Kienitz and Wetterau 2014). One first uses replication to determine the probability distributions of the component rates $\tilde{R}_1(t_{ex})$, $\tilde{R}_2(t_{ex})$ under the forward measure (see Hagan 2003, Andersen and Piterbarg 2010, Hagan *et al.* 2020). Then a copula is used to model the joint distribution of $\tilde{R}_1(t_{ex})$, $\tilde{R}(t_{ex})$ in a way that is consistent with the probability distributions of the individual rates $\tilde{R}_1(t_{ex})$ and $\tilde{R}_2(t_{ex})$. Finally, the value ot the CMS spread option is obtained by integrating to find the expected value of the payoff under this joint distribution.

There are several issues with this approach. First, replication values quadratic derivatives by replacing them with a portfolio of vanilla options with different strikes; as the spacing between strikes goes to zero, the portfolio's payoff converges to the quadratic payoff, and so the value of the portfolio converges to the value of the quadratic derivative. However, convergence requires extremely high strikes in the portfolio; often strikes in the range of 12–16%. At these strikes there is no liquid market for European options, so market prices are basically unknown. Convergence also requires densely spaced strikes. Even strikes spaced 5 bps apart requires valuing hundreds of swaptions, making this approach computationally intensive. Second, copulas focus on the joint distribution of $\tilde{R}_1$, $\tilde{R}_2$ and not the difference $\tilde{R}_1 - \tilde{R}_2$. Since $\tilde{R}_1$ and $\tilde{R}_2$ can be much larger than $\tilde{R}_1 - \tilde{R}_2$, small errors in $\tilde{R}_1$, $\tilde{R}_2$ lead to larger relative errors in $\tilde{R}_1 - \tilde{R}_2$. Finally, copulas are a mathematical description of a joint distribution, and not a physical model which leads to the joint distribution, so it is difficult to ascertain a copula's limits of applicability.

Of these issues, computational intensity has been the most challenging. Pricing a single callable range note based on CMS spreads may require thousands of European spread options. This was the original motivation for seeking a simpler alternative method of pricing European spread options.

2. Single rate CMS pricing

Let us briefly recap the standard convexity analysis (Hagan 2003, Andersen and Piterbarg 2010, Hagan *et al.* 2020) for CMS options on a singlequestionable = ko swap rate. Let $\tilde{R}_j(T)$ be either of the component swap rates $\tilde{R}_1(T)$ or $\tilde{R}_2(T)$, and let T_0 and T_{n_j} be the swap's start and end dates. Equating the fixed and floating leg of the swaps yields†

$$\tilde{R}_j(T) \sum_{i=1}^{n_j} \delta_i Z^i(T) = V^j_{flt}(T). \tag{5a}$$

Here,

$$Z^i(T) \equiv Z(T; T_i) \tag{5b}$$

is the zero coupon bond (discount factor) for the ith fixed leg pay date T_i at seen at date T, δ_i is the day count fraction (year fraction) for the swap's ith fixed leg interval, and $V^j_{flt}(T)$ is the value of the floating leg as seen at date T. The swap rate is then

$$\tilde{R}_j(T) = \frac{V^j_{flt}(T)/Z^0(T)}{L^j(T)}, \tag{6a}$$

where

$$L^j(T) = \frac{\sum_{i=1}^{n_j} \delta_i Z^i(T)}{Z^0(T)} \tag{6b}$$

is the forward level (PV01) forwarded to the swap's start date T_0, as seen at date T.

The payoffs of a CMS caplet, floorlet, and swaplet are

$$V^j_c(t_{ex}, K) = Z(t_{ex}; T_0)\left[\tilde{R}_j(t_{ex}) - K\right]_+, \tag{7a}$$

$$V^j_p(t_{ex}, K) = Z(t_{ex}; T_0)\left[K - \tilde{R}_j(t_{ex})\right]_+, \tag{7b}$$

$$V^j_s(t_{ex}, K) = Z(t_{ex}; T_0)\left[\tilde{R}_j(t_{ex}) - K\right], \tag{7c}$$

where K is the strike and t_{ex} is the expiry date. We first value the CMS caplet. We adopt the zero coupon bond $Z^0(T) \equiv$

† The value of the floating leg is often taken to be $Z(T; T_0) - Z(T; T_{n_j})$ in theoretical analyses, but this requires the forecast and discount curves to be identical. Currently forecast and discount rates differ by more than 25 bps in most currencies, so we avoid this assumption.

$Z(T;T_0)$ as our numeraire. Then the value of the CMS caplet at any earlier date t is

$$V_c^j(t,K) = Z(t;T_0)\,\mathrm{E}\left\{\left[\tilde{R}_j(t_{ex}) - K\right]_+ \,\middle|\, t\right\}, \tag{8}$$

where ‘E{}’ is the expected value under the forward measure and where ‘| t’ is shorthand for ‘given all the information known at time t’.

The swap rate $\tilde{R}_j(t_{ex})$ is not a Martingale in the forward measure, so the next step is to switch measures. The forward level $L^j(T)$ defined in (6b) is a Martingale in the forward measure, since it is a tradeable instrument (a portfolio of zero coupon bonds) divided by the numeraire $Z^0(T) \equiv Z(T;T_0)$. Consequently,

$$\mathrm{E}\left\{\frac{L^j(T)}{L^j(t)} \,\middle|\, t\right\} = 1. \tag{9}$$

This lets us define the annuity (Jamshidean) measure E^j by

$$\mathrm{E}^j\{A(T) \mid t\} \equiv \mathrm{E}\left\{A(T)\frac{L^j(T)}{L^j(t)} \,\middle|\, t\right\}. \tag{10}$$

It is easily seen that the swap rate $\tilde{R}_j$ is a Martingale in this measure: Since the floating leg is a tradeable instrument, $V_{flt}^j(T)/Z(T;T_0)$ is a Martingale, and so

$$\begin{aligned}\mathrm{E}^j\left\{\tilde{R}_j(T) \mid t\right\} &= \mathrm{E}\left\{\tilde{R}_j(T)\frac{L^j(T)}{L^j(t)} \,\middle|\, t\right\} \\ &= \frac{1}{L^j(t)}\mathrm{E}\left\{\frac{V_{flt}^j(T)}{Z(T;T_0)} \,\middle|\, t\right\} \\ &= \frac{V_{flt}^j(t)/Z^0(t)}{L^j(t)} = \tilde{R}_j^0,\end{aligned} \tag{11a}$$

where

$$\tilde{R}_j^0 \equiv R_j(t). \tag{11b}$$

We switch from the forward measure to the annuity measure by multiplying and dividing the payoff by $L^j(T)/L^j(t)$. This yields

$$V_c^j(t,K) = Z(t;T_0)\,\mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex}) - K\right]_+ \frac{L^j(t)}{L^j(t_{ex})} \,\middle|\, t\right\}. \tag{12}$$

Breaking up $L^j(t)/L^j(T) = 1 + (L^j(t)/L^j(T) - 1)$ yields

$$\begin{aligned}V_c^j(t,K) &= Z(t;T_0)\,\mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex}) - K\right]_+ \,\middle|\, t\right\} \\ &\quad + Z(t;T_0)\,\mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex}) - K\right]_+ \left(\frac{L^j(t)}{L^j(t_{ex})} - 1\right) \,\middle|\, t\right\}.\end{aligned} \tag{13}$$

The first term in $V_c^j(t,K)$ is a standard call option on an asset $\tilde{R}_j(T)$ which is a Martingale in the current measure. The second term is the convexity correction.

Convexity corrections arise because changes in the level $L^j(T)$ are correlated with changes in the swap rate $\tilde{R}_j(T)$. As rates rise, discounting increases which decreases the level $L^j(T)$. However, option prices depend on rates on the scale of $\sim \sigma_N t_{ex}^{1/2}$, where σ_N is the option's normal volatility, while the level $L^j(T)$ depends much more gradually on the rates. In fact, using parallel rate shifts shows that even if rates rise by 100 bps, the 5 year swap rate's level decreases by less than 2.5%, and that this change is nearly linear in the rate shift. Consequently, convexity corrections are small, and while they are too large to be neglected, they are too small to merit elaborate modeling. Accordingly, the next step is to make a simple model of the yield curve which relates the level $L^j(T)$ to the swap rate:

$$L^j(t)/L^j(T) = F\left(R^j(T)\right). \tag{14}$$

Different models for $F(R^j(t))$ are used in practice. Often parallel shifts of the yield curve are used to obtain $F(R^j(T))$, but some firms use simpler models. See Hagan (2003), Hagan *et al.* (2020) and Andersen and Piterbarg (2010). Regardless of the model used, one expands

$$\frac{L^j(t)}{L^j(T)} = 1 + M_j\left(\tilde{R}_j(T) - R_j^0\right) + \cdots, \tag{15}$$

where the *convexity coefficient* M_j is slightly different at different banks because of differing models. Substituting this expansion into (13), neglecting higher order terms, and replacing

$$\begin{aligned}&\mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex}) - K\right]_+ \left(\tilde{R}_j(T) - R_j^0\right) \,\middle|\, t\right\} \\ &\quad = \mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex}) - K\right]_+^2 \,\middle|\, t\right\} \\ &\qquad - \left(R_j^0 - K\right)\mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex}) - K\right]_+ \,\middle|\, t\right\},\end{aligned} \tag{16}$$

yields

$$\begin{aligned}V_c^j(t,K) &= Z(t;T_0)\left(1 - M_j\left(R_j^0 - K\right)\right) \\ &\quad \times \mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex}) - K\right]_+ \,\middle|\, t\right\} \\ &\quad + Z(t;T_0)\,M_j\mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex}) - K\right]_+^2 \,\middle|\, t\right\}.\end{aligned} \tag{17a}$$

Repeating the argument *mutandis mutandi* for CMS floorlets yields

$$\begin{aligned}V_p^j(t,K) &= Z(t;T_0)\left(1 + M_j\left(K - R_j^0\right)\right) \\ &\quad \times \mathrm{E}^j\left\{\left[K - \tilde{R}_j(t_{ex})\right]_+ \,\middle|\, t\right\} \\ &\quad - Z(t;T_0)\,M_j\mathrm{E}^j\left\{\left[K - \tilde{R}_j(t_{ex})\right]_+^2 \,\middle|\, t\right\}.\end{aligned} \tag{17b}$$

Similarly, the CMS swaplet value is

$$\begin{aligned}V_s^j(t,K) &= Z(t;T_0) \\ &\quad \times \left(R_j^0 - K + M_j\mathrm{E}^j\left\{\left(\tilde{R}_j(t_{ex}) - R_j^0\right)^2 \,\middle|\, t\right\}\right),\end{aligned} \tag{17c}$$

where we have used

$$\mathrm{E}^j\left\{\left(\tilde{R}_j(t_{ex})-K\right)^2\Big|\,t\right\}=\left(R_j^0-K\right)^2 + \mathrm{E}^j\left\{\left(\tilde{R}_j(t_{ex})-R_j^0\right)^2\Big|\,t\right\} \quad (18)$$

in the deriving the swaplet value.

2.1. Summary of single rate CMS prices

The CMS option prices can be written concisely as

$$V_c^j(t,K)=Z(t;T_0)\left\{\left(1-M_j\left(R_j^0-K\right)\right)C^j\left(t,R_j^0,K\right) + M_jQ_c^j\left(t,R_j^0,K\right)\right\} \quad (19a)$$

$$V_p^j(t,K)=Z(t;T_0)\left\{\left(1+M_j\left(K-R_j^0\right)\right)P^j\left(t,R_j^0,K\right) - M_jQ_p^j\left(t,R_j^0,K\right)\right\}, \quad (19b)$$

$$V_s^j(t,K)=Z(t;T_0)\left\{R_j^0-K+M_jQ_s^j\left(t,R_j^0\right)\right\}. \quad (19c)$$

Here

$$C^j\left(t,R_j^0,K\right)=\mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex})-K\right]_+\Big|\,t\right\}, \quad (20a)$$

$$P^j\left(t,R_j^0,K\right)=\mathrm{E}^j\left\{\left[K-\tilde{R}_j(t_{ex})\right]_+\Big|\,t\right\}, \quad (20b)$$

are canonical calls and puts, and

$$Q_c^j\left(t,R_j^0,K\right)=\mathrm{E}^j\left\{\left[\tilde{R}_j(t_{ex})-K\right]_+^2\Big|\,t\right\}, \quad (21a)$$

$$Q_p^j\left(t,R_j^0,K\right)=\mathrm{E}^j\left\{\left[K-\tilde{R}_j(t_{ex})\right]_+^2\Big|\,t\right\}, \quad (21b)$$

$$Q_s^j\left(t,R_j^0\right)=\mathrm{E}^j\left\{\left(\tilde{R}_j(t_{ex})-R_j^0\right)^2\Big|\,t\right\}, \quad (21c)$$

are canonical quadratic calls, puts, and swaplets, evaluated in a measure in which the underlying rate $\tilde{R}_j(T)$ is a Martingale.

Since the swap rate $\tilde{R}_j(T)$ is a Martingale in the measure $\mathrm{E}^j\{\}$, the calls $C^j(t,R_j^0,K)$ and puts $P^j(t,R_j^0,K)$ are standard vanilla swaptions, and can be valued using the desk's volatility model and marks for swaptions. Typically this would be a SABR or Heston model for $\tilde{R}_j(T)$. The quadratic options $Q_c^j(t,R_j^0,K)$, $Q_p^j(t,R_j^0,K)$, and $Q_s^j(t,R_j^0)$ are usually evaluated using the same volatility model, either by replication (Hagan 2003) or by using the SABR model's closed form formulas for quadratic options (Hagan *et al.* 2019, 2020).

Since the quadratic swap $Q_s^j(t,R_j^0)$ is independent of the strike K, the values of CMS swaplets are often expressed as

$$V_s^j(t,K)=Z(t;T_0)\left(R_j^0+\Delta R_j^{adj}-K\right), \quad (22a)$$

where the swap rate's convexity adjustment is

$$\Delta R_j^{adj}=M_j\,\mathrm{E}^j\left\{\left(\tilde{R}_j(t_{ex})-R_j^0\right)^2\Big|\,t\right\}. \quad (22b)$$

It is a not uncommon practice to price CMS caplets and floorlets by replacing the forward swap rate R_j^0 by the convexity adjusted swap rate $R_j^0+\Delta R_j^{adj}$, and simply ignoring the quadratic call and put options. I.e. by assuming that the only effect of convexity is to shift the forward swap rate. This is incorrect since the swap rate's convexity adjustment ΔR_j^{adj} is the result of averaging the quadratic $(\tilde{R}_j(t_{ex})-R_j^0)^2$ over all paths, whereas the quadratic calls and puts average the quadratic over only high rate or low rate paths.

2.2. Call-put parity

Since $\tilde{R}_j(T)$ is a Martingale in $\mathrm{E}^j\{\}$, call-put parity for vanilla options yields

$$C^j\left(t,R_j^0,K\right)-P^j\left(t,R_j^0,K\right)=\mathrm{E}^j\left\{\tilde{R}_j(t_{ex})-K\,\big|\,t\right\}=R_j^0-K. \quad (23a)$$

For quadratic options, the equivalent identity is

$$Q_c^j\left(t,R_j^0,K\right)+Q_p^j\left(t,R_j^0,K\right)=\mathrm{E}^j\left\{\left(\tilde{R}_j(t_{ex})-K\right)^2\Big|\,t\right\} = \left(R_j^0-K\right)^2+Q_s^j\left(t,R_j^0\right), \quad (23b)$$

where (18) has been used in obtaining this result. Consequently, CMS option values must satisfy the call-put parity relation

$$V_c^j(t,K)-V_p^j(t,K)=Z(t;T_0)\left\{R_j^0-K-M_jQ_s^j\left(R_j^0\right)\right\} = V_s^j(t,K), \quad (24)$$

regardless of the model used. The closed form fomulas for quadratic options under the SABR model in Hagan *et al.* (2019) are only accurate through $O(\varepsilon^2)$, but they satisfy call-put parity exactly. In particular,

$$Q_c^j\left(t,R_j^0,K\right)+Q_p^j\left(t,R_j^0,K\right)-\left(R_j^0-K\right)^2 \quad (25)$$

is exactly independent of K.

3. CMS options on spreads

We use the same approach to value options on CMS spreads,

$$\tilde{S}(t)\equiv\tilde{R}_1(t)-\tilde{R}_2(t), \quad (26)$$

creating a numeraire in which the spread $\tilde{S}(t)$ is a Martingale, analyzing the convexity correction, combining the volatility smile models for $\tilde{R}_1(T)$ and $\tilde{R}_2(T)$ to obtain a normal ($\beta=0$) SABR model for the spread $\tilde{S}(t)$, and finally using this model to evaluate the constituent vanilla and quadratic options.

3.1. Change of numeraire

We need to find a measure H in which

$$\mathrm{E}^H\left\{\tilde{S}(T)\Big|\,t\right\}=S^0, \tag{27}$$

where $S^0\equiv\tilde{S}(t)$. We cannot use $L^1(T)/L^1(t)$ or $L^2(T)/L^2(t)$ as the numeraire, since it would be surprising† if $\tilde{R}_1(t)$ were a Martingale in $L^2(T)/L^2(t)$ or $\tilde{R}_2(t)$ were a Martingale $L^1(T)/L^1(t)$. Instead we define a hybrid numeraire

$$H(T)=\gamma\frac{L^1(T)}{L^1(t)}+(1-\gamma)\frac{L^2(T)}{L^2(t)}, \tag{28}$$

where $\gamma(T)$ will be chosen later‡. Since both $L^1(T)$ and $L^2(T)$ are Martingales in the forward measure,

$$\mathrm{E}\{H(T)|\,t\}=H(t)\equiv 1, \tag{29}$$

so we can use $H(t)$ as a numeraire and define the measure

$$\mathrm{E}^H\{A(T)|\,t\}\equiv\mathrm{E}\{A(T)H(T)|\,t\}. \tag{30}$$

To make the spread $\tilde{S}(T)$ a Martingale, we need to choose $\gamma(T)$ so that (27) is satisfied. So

$$\begin{aligned}
&\mathrm{E}^H\left\{\tilde{S}(T)\Big|\,t\right\}\\
&\quad\equiv\mathrm{E}\left\{\left(\tilde{R}_1(T)-\tilde{R}_2(T)\right)\left(\gamma\frac{L^1(T)}{L^1(t)}+(1-\gamma)\frac{L^2(T)}{L^2(t)}\right)\Bigg|\,t\right\}\\
&\quad=R_1^0-R_2^0-\mathrm{E}\left\{\tilde{R}_1(T)\left(\frac{L^1(T)}{L^1(t)}-\frac{L^2(T)}{L^2(t)}\right)\Bigg|\,t\right\}\\
&\qquad+\gamma\mathrm{E}\left\{\left[\tilde{R}_1(T)-\tilde{R}_2(T)\right]\left(\frac{L^1(T)}{L^1(t)}-\frac{L^2(T)}{L^2(t)}\right)\Bigg|\,t\right\}.
\end{aligned} \tag{31}$$

Thus, we choose

$$\gamma(T)=\frac{\mathrm{E}\left\{\left[\tilde{R}_1(T)-R_1^0\right]\left(\frac{L^1(T)}{L^1(t)}-\frac{L^2(T)}{L^2(t)}\right)\Big|\,t\right\}}{\mathrm{E}\left\{\left[\tilde{S}(T)-S^0\right]\left(\frac{L^1(T)}{L^1(t)}-\frac{L^2(T)}{L^2(t)}\right)\Big|\,t\right\}}. \tag{32}$$

As yet, this prescription for γ is abstract. To make it concrete requires modeling the connections between the levels L^1, L^2 and the swap rates $\tilde{R}_1$, $\tilde{R}_2$.

† Under parallel shifts, the levels $L^1(T)$ and $L^2(T)$ shorten at different rates, so both $\tilde{R}_1$ and $\tilde{R}_2$ could be Martingales only under a very contrived set of yield curve shifts.
‡ Alternatively, $H(T)=\frac{\gamma L^1(T)+(1-\gamma)L^2(T)}{\gamma L^1(t)+(1-\gamma)L^2(t)}$ works equally well as the hybrid numeraire.

3.2. CMS spread convexity

Let us first consider the CMS caplet. From the payoff in (1a), the caplet value is

$$V_c^S(t,K)=Z(t;T_0)\,\mathrm{E}\left\{\left[\tilde{S}(t_{ex})-K\right]_+\Big|\,t\right\} \tag{33}$$

in the forward measure. We switch to the hybrid numeraire by multiplying and dividing by $H(t_{ex})$, obtaining

$$V_c^S(t,K)=Z(t;T_0)\,\mathrm{E}^H\left\{\frac{\left[\tilde{S}(t_{ex})-K\right]_+}{H(t_{ex})}\Bigg|\,t\right\}, \tag{34}$$

which we re-write as

$$\begin{aligned}
V_c^S(t,K)&=Z(t;T_0)\,\mathrm{E}^H\left\{\left[\tilde{S}(t_{ex})-K\right]_+\Big|\,t\right\}\\
&\quad+Z(t;T_0)\,\mathrm{E}^H\left\{\left[\tilde{S}(t_{ex})-K\right]_+\left(\frac{1}{H(t_{ex})}-1\right)\Bigg|\,t\right\}.
\end{aligned} \tag{35}$$

Since the spread $\tilde{S}(T)$ is a Martingale in the measure $\mathrm{E}^H\{\}$, the first term is a standard call option on the spread. The second term is the convexity correction.

Convexity corrections arise from the correlations of the spread with the levels L^j. As before, the levels only depend gradually on rates, so we expect convexity correction to be small. The next step is to create a simple model§ of the yield curve which relates the levels L^1, L^2 and the swap rates $\tilde{R}_1$, $\tilde{R}_2$. We need to allow L^j to depend on both $\tilde{R}_1(T)$ and $\tilde{R}_2(T)$,

$$L^1(T)/L^1(t)=F^1\left(\tilde{R}_1(T),\tilde{R}_2(T)\right), \tag{36a}$$

$$L^2(T)/L^2(t)=F^2\left(\tilde{R}_1(T),\tilde{R}_2(T)\right), \tag{36b}$$

since surely both swap PV01's change if either $\tilde{R}_1(T)$ or $\tilde{R}_2(T)$ changes. Suppose that we have created such a model.¶ Substituting the expansion

$$\begin{aligned}
L^1(T)/L^1(t)&=1-b_1^1\left(\tilde{R}_1(T)-R_1^0\right)\\
&\quad-b_2^1\left(\tilde{R}_2(T)-R_2^0\right)+\cdots,
\end{aligned} \tag{37a}$$

$$\begin{aligned}
L^2(T)/L^2(t)&=1-b_1^2\left(\tilde{R}_1(T)-R_1^0\right)\\
&\quad-b_2^2\left(\tilde{R}_2(T)-R_2^0\right)+\cdots
\end{aligned} \tag{37b}$$

§ As we shall see, we don't actually need to create such a model. Once the *form* of the convexity correction has been derived, the only effect of the model is to determine the value of the convexity coefficient M_s for the spread. Instead of determining this value from a model, it can be determined by requiring the CMS swaplet on the spread to be consistent with the CMS swaplets on the component swap rates $\tilde{R}_1(T),\tilde{R}_2(T)$.
¶ It suffices to use simple parallel shifts and tilts of the yield curve,

$$\frac{Z(T;T_i)}{Z(T;T_0)}=\frac{Z(t,T_i)}{Z(T;T_0)}e^{-\alpha(T_i-T_0)-\beta(T_i-T_0)^2}.$$

This defines both $\tilde{R}_1(T)$, $\tilde{R}_2(T)$ and $L^1(T)/L^1(t)$, $L^2(T)/L^2(t)$ in terms of α and β, and thus implicitly defines $F^{(1)}(\tilde{R}_1,\tilde{R}_2)$ and $F^{(2)}(\tilde{R}_1,\tilde{R}_2)$.

into (35) and neglecting all higher order terms yields

$$V_c^S(t,K) = Z(t;T_0)\,\mathrm{E}^H\left\{\left[\tilde S(t_{ex}) - K\right]_+ \Big|\, t\right\} + Z(t;T_0)\,\mathrm{E}^H\left\{\left[\tilde S(t_{ex}) - K\right]_+ \left(c_1\left[\tilde R_1(t_{ex}) - R_1^0\right] + c_2\left[\tilde R_2(t_{ex}) - R_2^0\right]\right)\Big|\, t\right\}, \tag{38a}$$

with

$$c_1 = \gamma b_1^1 + (1-\gamma)\,b_1^2, \tag{38b}$$

$$c_2 = \gamma b_2^1 + (1-\gamma)\,b_2^2. \tag{38c}$$

We decompose $c_1(\tilde R_1(t_{ex}) - R_1^0) + c_2(\tilde R_2(t_{ex}) - R_2^0)$ into a term linear in the spread $\tilde S(t_{ex}) - S_0$ and a term which is uncorrelated with the spread. To do this, let

$$v_{12} = \mathrm{E}^H\left\{\left(\tilde R_1(t_{ex}) - R_1^0\right)\left(\tilde R_2(t_{ex}) - R_2^0\right)\Big|\, t\right\} \tag{39}$$

be the covariances of $\tilde R_1(t_{ex}), \tilde R_2(t_{ex})$. The linear combination

$$\tilde R_\perp(t_{ex}) - R_\perp^0 \equiv \frac{(v_{22} - v_{12})\left(\tilde R_1(t_{ex}) - R_1^0\right) + (v_{11} - v_{12})\left(\tilde R_2(t_{ex}) - R_2^0\right)}{v_{11} - 2v_{12} + v_{22}} \tag{40}$$

is uncorrelated with the spread,

$$\mathrm{E}^H\left\{\left(\tilde S(t_{ex}) - S^0\right)\left(\tilde R_\perp(t_{ex}) - R_\perp^0\right)\Big|\, t\right\} = 0, \tag{41}$$

so we decompose

$$c_1\left(\tilde R_1(t_{ex}) - R_1^0\right) + c_2\left(\tilde R_2(t_{ex}) - R_2^0\right) = M_s\left(\tilde S(t_{ex}) - S^0\right) + M_\perp\left(\tilde R_\perp(t_{ex}) - R_\perp^0\right), \tag{42a}$$

with

$$M_s = \frac{c_1(v_{11} - v_{12}) - c_2(v_{22} - v_{12})}{v_{11} - 2v_{12} + v_{22}}, \tag{42b}$$

$$M_\perp = c_1 + c_2. \tag{42c}$$

With this decomposition, the caplet becomes

$$V_c^S(t,K) = Z(t;T_0)\,\mathrm{E}^H\left\{\left[\tilde S(t_{ex}) - K\right]_+ \Big|\, t\right\} + Z(t;T_0)\,M_s\mathrm{E}^H \times\left\{\left[\tilde S(t_{ex}) - K\right]_+ \left(\tilde S(t_{ex}) - S^0\right)\Big|\, t\right\} + Z(t;T_0)\,M_\perp\mathrm{E}^H \times\left\{\left[\tilde S(t_{ex}) - K\right]_+ \left(\tilde R_\perp(t_{ex}) - R_\perp^0\right)\Big|\, t\right\}. \tag{43}$$

Even though

$$\mathrm{E}^H\left\{\left(\tilde S(t_{ex}) - K'\right)\left(\tilde R_\perp(t_{ex}) - R_\perp^0\right)\Big|\, t\right\} = 0 \tag{44}$$

for any K', the last term isn't zero. It represents the 'convexity of the convexity ', and we expect it to be very small, of the same magnitude as the quadratic terms that we've already neglected in expanding the $F^j(\tilde R_1(T), \tilde R_2(T))$. Accordingly, we model the convexity correction by neglecting this term, obtaining

$$V_c^S(t,K) = Z(t;T_0)\,\mathrm{E}^H\left\{\left[\tilde S(t_{ex}) - K\right]_+ \Big|\, t\right\} + Z(t;T_0)\,M_s\mathrm{E}^H \times\left\{\left[\tilde S(t_{ex}) - K\right]_+ \left(\tilde S(t_{ex}) - S^0\right)\Big|\, t\right\}. \tag{45}$$

This is our model of the price of a CMS caplet on the spread $\tilde S(t_{ex})$, which we write more canonically as

$$V_c^S(t,K) = Z(t;T_0)\left(1 - M_s\left(S^0 - K\right)\right)\mathrm{E}^H \times\left\{\left[\tilde S(t_{ex}) - K\right]_+ \Big|\, t\right\} + Z(t;T_0)\,M_s\mathrm{E}^H\left\{\left[\tilde S(t_{ex}) - K\right]_+^2 \Big|\, t\right\}. \tag{46}$$

The prices of CMS floorlets and swaplets on the spread $\tilde S(t_{ex})$ can be obtained by repeating the above arguments *mutandis mutandi*. This yields

$$V_c^S(t,K) = Z(t;T_0)\left\{\left(1 - M_s\left(S^0 - K\right)\right)C^S\left(t,S^0,K\right) + M_sQ_c^s\left(t,S^0,K\right)\right\}, \tag{47a}$$

$$V_p^S(t,K) = Z(t;T_0)\left\{\left(1 + M_s\left(K - S^0\right)\right)P^S\left(t,S^0,K\right) - M_sQ_p^s\left(t,S^0,K\right)\right\}, \tag{47b}$$

$$V_s^S(t,K) = Z(t;T_0)\left\{S^0 - K + M_sQ_s^s\left(t,S^0\right)\right\}, \tag{47c}$$

where

$$C^S\left(t,S^0,K\right) = \mathrm{E}^H\left\{\left[\tilde S(t_{ex}) - K\right]_+ \Big|\, t\right\}, \tag{48a}$$

$$P^S\left(t,S^0,K\right) = \mathrm{E}^H\left\{\left[K - \tilde S(t_{ex})\right]_+ \Big|\, t\right\}, \tag{48b}$$

are the canonical calls and puts, and where

$$Q_c^S\left(t,S^0,K\right) = \mathrm{E}^H\left\{\left[\tilde S(t_{ex}) - K\right]_+^2 \Big|\, t\right\}, \tag{49a}$$

$$Q_p^S\left(t,S^0,K\right) = \mathrm{E}^H\left\{\left[K - \tilde S(t_{ex})\right]_+^2 \Big|\, t\right\}, \tag{49b}$$

$$Q_s^S\left(t,S^0\right) = \mathrm{E}^H\left\{\left(\tilde S(t_{ex}) - S^0\right)^2 \Big|\, t\right\}, \tag{49c}$$

are the canonical quadratic options. These formulas are identical to the fomulas (19a)–(21c) for CMS options on a single rate. To evaluate them we need to determine the convexity coefficient M_s and the volatility model for the spread $S(T)$.

Since the quadratic swap $Q_s^S(t,S^0)$ is independent of the strike K, the value of the CMS swaplet can be written in terms of a convexity-adjusted forward spread,

$$\frac{V_s^S(t,K)}{Z(t;T_0)} \equiv S^0 + \Delta S^{adj} - K = R_1^0 - R_2^0 + \Delta S^{adj} - K, \tag{50a}$$

where the spread's convexity adjustment is

$$\Delta S^{adj} = M_s Q_s^S\left(t,S^0\right) \equiv M_s \mathrm{E}^H\left\{\left(\tilde{S}(t_{ex}) - S^0\right)^2 \Big| t\right\}. \tag{50b}$$

As with CMS options on a single swap rate, one cannot price CMS caplets and floorlets by simply replacing the forward spread S^0 by the convexity adjusted spread $S^0 + \Delta S^{adj}$. Instead one needs to properly include the quadratic call and put options.

3.2.1. The convexity coefficent M_s. The next step is to model the levels $L^1(T)$, $L^2(T)$ in terms of the swap rates $\tilde{R}_1(T)$, $\tilde{R}_2(T)$. Equations (36a)–(46) above would then determine the convexity coefficient M_s. However, we need these models to be consistent with the pricing of CMS options on the individual swap rates $\tilde{R}_1(T)$, $\tilde{R}_2(T)$, since this would let the desk use single rate CMS options to hedge the spread option. This consistency requirement will give us M_s directly, obviating the need to actually model L^1, L^2 in terms of $\tilde{R}_1$, $\tilde{R}_2$.

In the forward measure the value of the CMS swaplet on the spread is

$$\begin{aligned}\frac{V_s^S(t,K)}{Z(t;T_0)} &= \mathrm{E}\left\{\tilde{S}(t_{ex}) - K \Big| t\right\}\\ &= \mathrm{E}\left\{\tilde{R}_1(t_{ex})\big| t\right\} - \mathrm{E}\left\{\tilde{R}_2(t_{ex})\big| t\right\} - K.\end{aligned} \tag{51}$$

Comparing this with (50a) shows that

$$\mathrm{E}\left\{\tilde{R}_1(t_{ex})\big| t\right\} - \mathrm{E}\left\{\tilde{R}_2(t_{ex})\big| t\right\} = R_1^0 - R_2^0 + \Delta S^{adj}. \tag{52}$$

Similarly, the value of the CMS swaplet on the rate $\tilde{R}_j(T)$ is

$$\begin{aligned}\frac{V_s^j(t,K)}{Z(t;T_0)} &= \mathrm{E}\left\{\left[\tilde{R}_j(t_{ex}) - K\right]\big| t\right\}\\ &= R_j^0 + \Delta R_j^{adj} - K,\end{aligned} \tag{53}$$

so in the forward measure,

$$\mathrm{E}\left\{\tilde{R}_j(t_{ex})\big| t\right\} = R_j^0 + \Delta R_j^{adj}. \tag{54}$$

Clearly, for CMS spread options to be consistent with CMS options on the rates $\tilde{R}_1(T)$, $\tilde{R}_2(T)$, we must require

$$\Delta S^{adj} = \Delta R_1^{adj} - \Delta R_2^{adj}. \tag{55}$$

That is, consistency requires choosing

$$M_s = \frac{\Delta R_1^{adj} - \Delta R_2^{adj}}{Q_s^S\left(t,S^0\right)} = \frac{M_1 Q_s^1\left(t,R_1^0\right) - M_2 Q_s^2\left(t,R_2^0\right)}{Q_s^S\left(t,S^0\right)}. \tag{56}$$

Equation (56) is how we choose the convexity coefficient M_s. It is then consistent with the CMS convexity corrections of the component swap rates, regardless of how one chooses to model these single rate options. This means that in modeling the relationships $F(\tilde{R}^j(T))$ between single swap rates $\tilde{R}_j(T)$ and their levels $L^j(T)$, one can craft the models to those particular swaps. One is not forced to extend the model to simultaneously handle a second swap, which would degrade the model's handling of the first swap. In addition, when one models both swap rates together, then an option on the 30yr–5yr spread will model the 5yr rate differently from the 5yr rate in an option on the 5yr–1yr spread, which will be modeled differently from the 5yr rate in a CMS option on the 5yr rate. This would limit the ability of the various 5yr risks to offset each other, leading to noisier hedging. Since these options can be very long dated, even a small amount of hedging 'chatter' can be pernicious.

3.2.2. Call-put parity. Since $\tilde{S}(t_{ex})$ is a Martingale in the hybrid measure, we have call-put parity for the vanilla options,

$$C^S\left(t,S^0,K\right) - P^S\left(t,S^0,K\right) = \mathrm{E}^H\left\{\tilde{S}(t_{ex}) - K\Big| t\right\} = S^0 - K. \tag{57a}$$

For the canonical quadratic options, the equivalent identity is

$$\begin{aligned}Q_c^S\left(t,S^0,K\right) + Q_p^S\left(t,S^0,K\right) &= \mathrm{E}^H\left\{\left(\tilde{S}(t_{ex}) - K\right)^2\Big| t\right\}\\ &= \left(S^0 - K\right)^2 + Q_s^S\left(t,S^0\right).\end{aligned} \tag{57b}$$

Consequently, CMS spread options satisfy call-put parity

$$\begin{aligned}V_c^S(t,K) - V_p^S(t,K) &= Z(t;T_0)\left\{S^0 - K - M_s Q_s^S\left(t,S^0\right)\right\}\\ &= V_s^S(t,K).\end{aligned} \tag{58}$$

3.3. SABR model for the spread

It remains to derive the SABR model for $\tilde{S}(T)$, and then use this model to evaluate the canonical options explicitly. Without loss of generality we take today to be $t = 0$.

We assume that the individual swap rates $\tilde{R}_j$ can be modeled by the normal ($\beta = 0$) SABR model. Since swaptions are often modeled with $\beta \neq 0$, this may require using a shifter to convert the SABR model to $\beta = 0$. See Appendix 3. Since neither swap rate is a Martingale in this measure,

$$\mathrm{d}\tilde{R}_j = \varepsilon \tilde{A}_j \mathrm{d}W_j + \text{drift terms}, \tag{59a}$$

$$\mathrm{d}\tilde{A}_j = \varepsilon \nu_j \tilde{A}_j \mathrm{d}Z_j, \tag{59b}$$

$$\mathrm{d}W_j \mathrm{d}Z_j = \rho_{jj}\mathrm{d}T, \tag{59c}$$

for $j = 1, 2$. Consequently, the spread must satisfy

$$\mathrm{d}\tilde{S} = \varepsilon \tilde{A}_1 \mathrm{d}W_1 - \tilde{A}_2 \mathrm{d}W_2, \tag{60a}$$

$$\mathrm{d}\tilde{A}_1 = \varepsilon \nu_1 \tilde{A}_1 \mathrm{d}Z_1, \tag{60b}$$

$$\mathrm{d}\tilde{A}_2 = \varepsilon \nu_2 \tilde{A}_2 \mathrm{d}Z_2, \tag{60c}$$

since the drift terms cancel out exactly as the spread $\tilde{S}(T)$ is a Martingale in the hybrid measure.

The SABR parameters $\alpha_1 \equiv \tilde{A}_1(0)$, ρ_{11}, ν_1 and $\alpha_2 \equiv \tilde{A}_2(0)$, ρ_{22}, ν_2 for these models can be obtained by calibrating the theoretical SABR prices to the observed market smile for swaptions on $\tilde{R}_1$ and $\tilde{R}_2$. The correlation between the two swap rates,

$$\mathrm{d}W_1\mathrm{d}W_2 = c\,\mathrm{d}T, \tag{61a}$$

is also a critical parameter. To complete the model we need the remaining correlations,

$$\mathrm{d}W_1\,\mathrm{d}Z_2 = \rho_{12}\,\mathrm{d}T, \quad \mathrm{d}W_2\,\mathrm{d}Z_1 = \rho_{21}\,\mathrm{d}T, \quad \mathrm{d}Z_1\,\mathrm{d}Z_2 = r\,\mathrm{d}T. \tag{61b}$$

The correlations ρ_{12}, ρ_{21}, and r are not easily observable. Obtaining them by fitting the spread's volatility smile to market data would grossly overfit the data. However, swap rate volatilities are usually highly correlated, with swap rates becoming volatile or quiescent together, or, equivalently, with swaption prices rising and falling together. Accordingly, a common approach in practice is to ignore idiosyncrasies in the individual swaption volatilities, and model the FX volatilities as being driven by the same process,

$$\mathrm{d}Z_1 = \mathrm{d}Z_2 \equiv \mathrm{d}Z. \tag{62a}$$

This is the *single stochastic vol* model. It follows that

$$\rho_{12} = \rho_{11}, \quad \rho_{21} = \rho_{22}, \quad r \equiv 1. \tag{62b}$$

The advantage of the single stochastic vol model is that the volatility smile of the spread $\tilde{S}(T)$ is determined by a single extra parameter, the correlation c between $\tilde{R}_1$ and $\tilde{R}_2$. The other parameters are obtained directly from the SABR models for the component swaptions. So if we can fit c to, say, the spread swaption's at-the-money volatility, the model then determines the rest of the spread's volatility smile. This allows a desk to handle all the vega risks and hedges on the spread self-consistently.

We first derive the normal SABR model for $\tilde{S}(T)$ for the general correlations in (61a) and (61b), and then specialize it to the single stochastic vol model of (62a) and (62b).

3.3.1. The effective SABR parameters. In Appendix 1 we analyze European options on a basket

$$\tilde{S} = \sum_{j=1}^{n} \lambda_j \tilde{R}_j, \tag{63}$$

where each of the components $\tilde{R}_j$ is governed by a normal SABR model (59a) and (59b). Clearly,

$$\mathrm{d}\tilde{S} = \varepsilon \sum_{j=1}^{n} \lambda_j \tilde{A}_j\,\mathrm{d}W_j, \tag{64a}$$

$$\mathrm{d}\tilde{A}_j = \varepsilon \nu_j \tilde{A}_j \mathrm{d}Z_j \quad \text{for } j = 1, \dots n \tag{64b}$$

in the appropriate numeraire. Initially we consider the general correlation structure,

$$\mathrm{d}W_i\mathrm{d}W_j = c_{ij}\mathrm{d}T, \quad \mathrm{d}W_i\mathrm{d}Z_j = \rho_{ij}\mathrm{d}T, \quad \mathrm{d}Z_i\mathrm{d}Z_j = r_{ij}\mathrm{d}T. \tag{64c}$$

We use singular perturbation techniques to show that the probability density for $\tilde{S}(t_{ex})$, and thus European option prices, are identical to the probability density obtained from the effective SABR model

$$\mathrm{d}\tilde{S} = \varepsilon \tilde{A}\mathrm{d}W, \tag{65a}$$

$$\mathrm{d}\tilde{A} = \varepsilon \nu_{eff} \tilde{A}\mathrm{d}Z, \tag{65b}$$

$$\mathrm{d}W\,\mathrm{d}Z = \rho_{eff}\,\mathrm{d}T, \tag{65c}$$

where the effective SABR parameters $\alpha_{eff} = \tilde{A}(0)$, ρ_{eff}, and ν_{eff} are given by (A39a)–(A39c) with (A33b)–(A33f). These results are not exact, but they are accurate through $O(\varepsilon^2)$, the same accuracy as the original SABR formulas (Hagan *et al.* 2002). Besides options on spreads, these results can be applied to European options on baskets and on assets driven by multiple sources of noise.

In Appendix 2 we set $n = 2$ with $\lambda_1 = 1$ and $\lambda_2 = -1$ to obtain the SABR parameters $\alpha_{eff} \equiv \tilde{A}(0)$, ρ_{eff}, ν_{eff} for European options on spreads. The SABR parameters are first derived for general correlations (61a) and (61b) and are given by (A59a)–(A59c) and (A60a)–(A61c). These results are then specialized to the case of a single stochastic vol, $\mathrm{d}Z_1 = \mathrm{d}Z_2 \equiv \mathrm{d}Z$, where the correlations are given by (62b). For this case the SABR parameters are†

$$\alpha_{eff} = \Delta e^{\frac{1}{4}\varepsilon^2 \Gamma t_{ex}}, \tag{66a}$$

$$\rho_{eff} = \bar{\rho}\bar{\nu}/\sqrt{\bar{\nu}^2 + \kappa}, \tag{66b}$$

$$\nu_{eff} = \sqrt{\bar{\nu}^2 + \kappa}, \tag{66c}$$

where

$$\Delta^2(\boldsymbol{\alpha}) = \alpha_1^2 - 2c\alpha_1\alpha_2 + \alpha_2^2, \tag{67a}$$

$$\bar{\rho}(\boldsymbol{\alpha}) = \frac{\alpha_1\rho_{11} - \alpha_2\rho_{22}}{\Delta(\boldsymbol{\alpha})}, \tag{67b}$$

$$\bar{\nu}(\boldsymbol{\alpha}) = \frac{\alpha_1^2\nu_1 - c\alpha_1\alpha_2(\nu_1 + \nu_2) + \alpha_2^2\nu_2}{\Delta^2}, \tag{67c}$$

and

$$\Gamma = \left(1 - c^2\right)\frac{\alpha_1^2\alpha_2^2}{\Delta^4}(\nu_1 - \nu_2)^2 - \kappa(\boldsymbol{\alpha}), \tag{67d}$$

$$\begin{aligned}\kappa(\boldsymbol{\alpha}) = &\left\{3\left(1 - c^2\right)\frac{\alpha_1^2\alpha_2^2}{\Delta^4} - c\frac{\alpha_1\alpha_2}{\Delta^2}\right\}\bar{\rho}^2(\nu_1 - \nu_2)^2 \\ &+ \frac{\alpha_1\alpha_2}{\Delta^3}\{\alpha_1(\rho_{22} - c\rho_{11})\nu_1 \\ &+ \alpha_2(\rho_{11} - c\rho_{22})\nu_2\}\bar{\rho}(\nu_1 - \nu_2).\end{aligned} \tag{67e}$$

See (A63a)–(A65b). Here $\bar{\rho}$ and $\bar{\nu}$ represent the volatility weighted correlation and vol-of-vol, while Γ and κ arise from the dispersion of vols. Often Γ and κ are much smaller than $\bar{\nu}^2$. In this case we can omit these terms, and take $\rho^{eff} \simeq \bar{\rho}$, $\nu^{eff} \simeq \bar{\nu}$, $\Gamma \simeq 0$, at least for qualitative purposes. The effective SABR coefficients are then just Δ, $\bar{\rho}$, and $\bar{\nu}$.

† Since today is $t = 0$, t_{ex} is the time-to-expiry. Note that each different t_{ex} requires a slightly different α^{eff} to obtain the correct European option prices for that expiry date. Alternatively, if we replace $\mathrm{d}\tilde{S} = \varepsilon\tilde{A}\,\mathrm{d}W$ in the SABR model with $d\tilde{S} = \varepsilon\tilde{A}e^{\frac{1}{2}\varepsilon^2\Gamma T}\,\mathrm{d}W$, we can use $\alpha^{eff} = \Delta$ for all expiries at the cost of having a time dependent coefficient in the SABR model.

3.4. Explicit formulas

3.4.1. Option valuation under the normal model. The last step is to use the normal SABR model to find explicit formulas for valuing the canonical options. We can express these values more clearly if we first obtain their values under the normal model,

$$\mathrm{d}\tilde{S} = \sigma_N \mathrm{d}W. \tag{68}$$

Under the normal model, the values of vanilla call and put are†

$$C^S\left(0, S^0, K\right) \equiv (S_0 - K)\,\mathfrak{N}\left(\frac{S_0 - K}{\sigma_N\sqrt{t_{ex}}}\right) + \sigma_N\sqrt{t_{ex}}G\left(\frac{S_0 - K}{\sigma_N\sqrt{t_{ex}}}\right), \tag{69a}$$

$$P^S\left(0, S^0, K\right) \equiv (K - S_0)\,\mathfrak{N}\left(\frac{K - S_0}{\sigma_N\sqrt{t_{ex}}}\right) + \sigma_N\sqrt{t_{ex}}G\left(\frac{K - S_0}{\sigma_N\sqrt{t_{ex}}}\right), \tag{69b}$$

where $\mathfrak{N}()$ is the cumulative normal distribution and $G()$ is the Gaussian density. The values of the canonical quadratic options work out to be (Hagan *et al.* 2019)

$$Q_c^S\left(0, S_0, K\right) \equiv \left\{(S_0 - K)^2 + \sigma_N^2 t_{ex}\right\}\mathfrak{N}\left(\frac{S_0 - K}{\sigma_N\sqrt{t_{ex}}}\right) + (S_0 - K)\,\sigma_N\sqrt{t_{ex}}G\left(\frac{S_0 - K}{\sigma_N\sqrt{t_{ex}}}\right), \tag{70a}$$

$$Q_p^S\left(0, S_0, K\right) \equiv \left\{(S_0 - K)^2 + \sigma_N^2 t_{ex}\right\}\mathfrak{N}\left(\frac{K - S_0}{\sigma_N\sqrt{t_{ex}}}\right) + (K - S_0)\,\sigma_N\sqrt{t_{ex}}G\left(\frac{K - S_0}{\sigma_N\sqrt{t_{ex}}}\right), \tag{70b}$$

$$Q_s^S\left(0, S_0\right) \equiv \sigma_N^2 t_{ex}, \tag{70c}$$

3.4.2. Option values under the (normal) SABR model. Singular perturbation methods have been used to analyze vanilla options under the SABR model in Hagan *et al.* (2019). For the normal ($\beta = 0$) SABR model, this analysis shows that the vanilla option values are given by (69a) and (69b) with the implied (normal) volatility,

$$\sigma_N\left(K\right) = \sigma_N^{atm}\left(S_0\right)\cdot\frac{\zeta}{Y\left(\zeta\right)}, \tag{71a}$$

where the at-the-money volatility is

$$\sigma_N^{atm}\left(S_0\right) = \varepsilon\alpha_{eff}\left\{1 + \varepsilon^2\tfrac{2-3\rho_{eff}^2}{24}\nu_{eff}^2 t_{ex}\right\}, \tag{71b}$$

and

$$\zeta\left(K\right) = \frac{\nu_{eff}}{\alpha_{eff}}\left(S_0 - K\right), \tag{71c}$$

$$Y\left(\zeta\right) \equiv \ln\left(\frac{\sqrt{1 - 2\rho_{eff}\zeta + \zeta^2} - \rho_{eff} + \zeta}{1 - \rho_{eff}}\right). \tag{71d}$$

† The expiry date t_{ex} is also the time-to-expiry as we are taking today to be $t = 0$.

This formula is not exact, but it is accurate through $O(\varepsilon^2)$. There are many other slightly different formulas in use for the implied volatility $\sigma_N(K)$ under the SABR model. See Hagan *et al.* (2002) and Hagan *et al.* (2016) and the references cited within. These formulas are all accurate through $O(\varepsilon^2)$, but differences at higher order lead to perceived superiority for some variants. We believe that the above formula balances accuracy, stability, and simplicity, but see Hagan *et al.* (2016) for other formulas.

The values of quadratic calls, puts, and swaps under the SABR model are derived in Hagan *et al.* (2019). There it is found that

$$Q_c^S\left(0, S^0, K\right) \equiv \left\{(S_0 - K)^2 + s_Q^2 t_{ex}\right\}\mathfrak{N}\left(\frac{S_0^{adj} - K}{\sigma_Q\sqrt{t_{ex}}}\right) + \left(S_0^{adj} - K\right)\sigma_Q\sqrt{t_{ex}}G\left(\frac{S_0^{adj} - K}{\sigma_Q\sqrt{t_{ex}}}\right), \tag{72a}$$

$$Q_p^S\left(0, S^0, K\right) \equiv \left\{(S_0 - K)^2 + s_Q^2 t_{ex}\right\}\mathfrak{N}\left(\frac{K - S_0^{adj}}{\sigma_Q\sqrt{t_{ex}}}\right) + \left(K - S_0^{adj}\right)\sigma_Q\sqrt{t_{ex}}G\left(\frac{K - S_0^{adj}}{\sigma_Q\sqrt{t_{ex}}}\right), \tag{72b}$$

$$Q_s^S\left(0, S^0\right) \equiv (S_0 - K)^2 + s_Q^2 t_{ex}, \tag{72c}$$

through $O(\varepsilon^2)$. For the normal ($\beta = 0$) SABR model, the effective normal vol for quadratic options is

$$\sigma_Q\left(K\right) = \sigma_N\left(K\right)\left\{1 + \tfrac{1}{6}\varepsilon^2\nu_{eff}^2 t_{ex}\right\}, \tag{73a}$$

the quadratic swap vol is

$$s_Q = \sigma_N\left(S_0\right)\left\{1 + \varepsilon^2\tfrac{4+3\rho^2}{24}\nu_{eff}^2 t_{ex}\right\}, \tag{73b}$$

and the adjusted forward is

$$S_0^{adj}\left(K\right) = S_0 + \tfrac{1}{2}\frac{\sigma_Q^2\left(K\right) - \sigma_Q^2\left(S_0\right)}{K - S_0}t_{ex}. \tag{73c}$$

These formulas follow the same template as values of quadratic options under the normal model with $\sigma_Q(K)$, s_Q, and $S_0^{adj}(K)$ chosen to give the correct values under the normal SABR model. For compatibility, these quantities have been written in terms of the implied normal vols $\sigma_N(K)$, $\sigma_N(S_0)$ for vanilla swaptions. This means that regardless of the model used for the vanilla swaptions, the quadratic option values in (72a)–(73c) are accurate to within $O(\varepsilon^2)$ provided that the vanilla option volatilities $\sigma_N(K)$, $\sigma_N(S_0)$ are accurate to within $O(\varepsilon^2)$.

The need to adjust the spread $\tilde{S}(T)$ is surprising since $\tilde{S}(T)$ is a Martingale. Rewriting this adjusted spread as

$$S_0^{adj}\left(K\right) = S_0 + \tfrac{1}{2}\left(\frac{1}{K - S_0}\int_{S_0}^{K}\frac{d\sigma_Q^2\left(S'\right)}{dS'}dS'\right)t_{ex} \tag{74}$$

shows that even though $\tilde{S}(T)$ is a Martingale, as far as quadratic options are concerned, $\tilde{S}(T)$ advects with a velocity of $\frac{1}{2}d\sigma_Q^2(S)/dS$; the spread adjustment $S_0^{adj}(K)$ just represents the average advection velocity between S_0 and K. In a physical system this would represent thermophoresis (negative thermodiffusion), where some chemical species preferentially diffuse towards regions of higher diffusivity/temperature.

Note that the above quadratic pricing formulas (72a)–(73c) satisfy call put parity exactly since

$$\begin{aligned} Q_c^S\left(0,S^0,K\right)+Q_p^S\left(0,S^0,K\right) &= (S_0-K)^2+s_Q^2t_{ex} \\ &= Q_s^S\left(0.S^0\right). \end{aligned} \tag{75}$$

In particular,

$$Q_c^S\left(0,S^0,K\right)+Q_p^S\left(0,S^0,K\right)-\left(S^0-K\right)^2 \tag{76}$$

is exactly independent of K.

4. Summary and conclusions

We developed a method for explicitly pricing caplet, floorlet, and swaplets on CMS spreads,

$$\tilde{S}\left(T\right)=\tilde{R}_1\left(T\right)-\tilde{R}_2\left(T\right). \tag{77}$$

By analyzing and modeling the convexity correction, we found that these option values could be written as

$$\begin{aligned} V_c^S\left(0,K\right) &= Z\left(0;T_0\right)\left\{\left(1-M_s\left(S^0-K\right)\right)C^S\left(0,S^0,K\right)\right. \\ &\quad \left.+\ M_sQ_c^s\left(0,S^0,K\right)\right\}, \end{aligned} \tag{78a}$$

$$\begin{aligned} V_p^S\left(0,K\right) &= Z\left(0;T_0\right)\left\{\left(1+M_s\left(K-S^0\right)\right)P^S\left(0,S^0,K\right)\right. \\ &\quad \left.-\ M_sQ_p^s\left(0,S^0,K\right)\right\}, \end{aligned} \tag{78b}$$

$$V_s^S\left(0,K\right)=Z\left(0;T_0\right)\left\{S^0-K+M_sQ_s^s\left(0,S^0\right)\right\}, \tag{78c}$$

where M_s is the convexity coefficient, where $C^S(0,S^0,K)$ and $P^S(0,S^0,K)$ are canonical calls and puts, and where $Q_c^S(0,S^0,K)$, $Q_p^S(0,S^0,K)$, and $Q_s^S(0,S^0)$ are the canonical quadratic options. See equations (47a)–(49c).

Instead of using modeling to determine the convexity coefficient M_s, we choose M_s so that the CMS spread options are consistent with CMS options on the component swap rates $\tilde{R}_1(T)$, $\tilde{R}_2(T)$. This requires choosing M_s as

$$M_s=\frac{\Delta R_1^{adj}-\Delta R_2^{adj}}{Q_s^S\left(0,S^0\right)}=\frac{M_1Q_s^1\left(0,R_1^0\right)-M_2Q_s^2\left(0,R_2^0\right)}{Q_s^S\left(0,S^0\right)}, \tag{79}$$

where ΔR_1^{adj}, ΔR_2^{adj} are the convexity adjustments for the components swap rates $\tilde{R}_1(T)$, $\tilde{R}_2(T)$. See Equations (51)–(56).

To value the canonical options, we assumed the swap rates $\tilde{R}_1(T)$, $\tilde{R}_2(T)$ are modeled by normal SABR models, using a shifter if needed to make it so. See Appendix 3. Then

$$\mathrm{d}\tilde{R}_j=\varepsilon\tilde{A}_j\,\mathrm{d}W_j+\text{drift terms}, \tag{80a}$$

$$\mathrm{d}\tilde{A}_j=\varepsilon\nu_j\tilde{A}_j\,\mathrm{d}Z_j, \tag{80b}$$

with the correlations

$$\mathrm{d}W_1\mathrm{d}W_2=c\,\mathrm{d}T,\quad \mathrm{d}W_i\mathrm{d}Z_j=\rho_{ij}\mathrm{d}T,\quad \mathrm{d}Z_1\mathrm{d}Z_2=r\,\mathrm{d}T. \tag{80c}$$

We used singular perturbation analysis to combine these models into a normal SABR model for the spread:

$$\mathrm{d}\tilde{S}=\varepsilon\tilde{A}\,\mathrm{d}W, \tag{81a}$$

$$\mathrm{d}\tilde{A}=\varepsilon\nu_{eff}\tilde{A}\,\mathrm{d}Z, \tag{81b}$$

$$\mathrm{d}W\,\mathrm{d}Z=\rho_{eff}\,\mathrm{d}T. \tag{81c}$$

The SABR parameters $\alpha_{eff}\equiv\tilde{A}(0)$, ρ_{eff}, ν_{eff} for the spread model are given by (A59a)–(A59c) with (A60a)–(A61c).

The volatility smiles of the swaptions on $\tilde{R}_1(T)$ and $\tilde{R}_2(T)$ only determine the SABR parameters $\alpha_1\equiv A_1(0)$, ρ_{11}, ν_1 and $\alpha_2\equiv A_2(0)$, ρ_{22}, ν_2. This still leaves the rate-to-rate correlation c, the cross correlations ρ_{12}, ρ_{21}, and the vol-to-vol correlation r. Our favored way of setting these extra parameters is by using the single stochastic vol model, where it is assumed that $dZ_1\equiv dZ_2\equiv dZ$ so that $\rho_{12}=\rho_{11}$, $\rho_{21}=\rho_{22}$, and $r=1$. In this case the spread only has a single extra parameter, the rate-to-rate correlation c, and the SABR parameters $\alpha_{eff}\equiv\tilde{A}(0)$, ρ_{eff}, ν_{eff} for the spread simplify to the expressions in (66a)–(66c) with (67a)–(67e). In this case the new parameter c can be found by calibrating the model to the at-the-money spread option.

Finally, under the SABR model, the values of the standard call and puts $C^S(0,S^0,K)$, $P^S(0,S^0,K)$ are given by (69a) and (69b) with the implied normal volatility (71a), (71d), and the quadratic option prices are given by (72a) and (73c).

Advantages of this approach are that it yields explicit formulas for the CMS spread options, making valuation trivial, it is fully consistent with CMS options on the component swap rates, and it needs only a single extra parameter, the rate-to-rate correlation c, that isn't determined by the SABR models for the component swap rates.

We also derived the effective SABR model on a basket of assets,

$$\tilde{S}\left(T\right)=\sum_{j=1}^{n}\lambda_j\tilde{R}_j\left(T\right), \tag{82}$$

assuming that each of the components in the basket is governed by a normal SABR model. This allows us to price European options on baskets and on assets driven by multiple sources of noise.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendices

Appendix 1. Derivation of the SABR model

We derive the effective SABR model for a basket of n underlying assets,

$$\tilde{S} = \sum_{j=1}^{n} \lambda_j \tilde{R}_j, \tag{A1}$$

where λ_j is the basket weight of asset j. Then in Appendix B we specialize the results to spreads by setting $n = 2$ with $\lambda_1 = 1$ and $\lambda_2 = -1$. We assume that each asset's forward price is governed by the *normal* ($\beta = 0$) SABR model, (59a) and (59b). Then

$$\mathrm{d}\tilde{S} = \varepsilon \sum_{j=1}^{n} \lambda_j \tilde{A}_j \mathrm{d}W_j, \tag{A2a}$$

$$\mathrm{d}\tilde{A}_j = \varepsilon \nu_j \tilde{A}_j \mathrm{d}Z_j \quad \text{for } j = 1, \ldots n, \tag{A2b}$$

in the appropriate measure, where we initially consider the full correlation structure

$$\mathrm{d}W_i \mathrm{d}W_j = c_{ij}\mathrm{d}T, \quad \mathrm{d}W_i \mathrm{d}Z_j = \rho_{ij}\mathrm{d}T, \quad \mathrm{d}Z_i \mathrm{d}Z_j = r_{ij}\mathrm{d}T. \tag{A2c}$$

The general correlation structure in (A2c) introduces too many unobservable parameters to be practical. So after obtaining the general results, we specialize to specific correlation structures which have a more parsimonious parameter set.

A.1. Conservation law

We derive the effective SABR model for $\tilde{S}(t_{ex})$ by following the approach in Hagan *et al.* (2013). We first obtain the effective forward equation that determines the probability density of $\tilde{S}(t_{ex})$, and then find the SABR model which generates the same effective forward equation, and thus the same probability distribution for $\tilde{S}(t_{ex})$ and prices of European option prices on $\tilde{S}(t_{ex})$.

If the volatilities weren't stochastic, so the $\tilde{A}_j$'s were constant, then we could combine the Brownian motions in (A2a) into a single Brownian motion

$$\mathrm{d}\tilde{S} = \varepsilon \Delta(\tilde{\mathbf{A}})\, \mathrm{d}W \tag{A3a}$$

where

$$\Delta(\tilde{\mathbf{A}}) = \left(\sum_{i,j} c_{ij} \lambda_i \lambda_j \tilde{A}_i \tilde{A}_j \right)^{1/2}. \tag{A3b}$$

With stochastic $\tilde{A}_j$'s, we still expect $\Delta(\tilde{\mathbf{A}})$ to be the leading order normal volatility for $\tilde{S}$.

Define the probability density

$$p(t, s, \boldsymbol{\alpha}; T, S, \mathbf{A}) = \mathrm{E}\left\{ \delta\left(\tilde{S}(T) - S\right) \delta^n \left(\tilde{\mathbf{A}}(T) - \mathbf{A}\right) \middle| \tilde{S}(t) = s, \tilde{\mathbf{A}}(t) = \boldsymbol{\alpha} \right\}, \tag{A4}$$

and the moments

$$\begin{aligned} Q^{(k)}(t, s, \boldsymbol{\alpha}; T, S) &= \mathrm{E}\left\{ \Delta^k(\tilde{\mathbf{A}}(T)) \delta\left(\tilde{S}(T) - S\right) \middle| \tilde{S}(t) = s, \tilde{\mathbf{A}}(t) = \boldsymbol{\alpha} \right\} \\ &= \int \cdots \int \Delta^k(\mathbf{A}) p(t, s, \boldsymbol{\alpha}; T, S, \mathbf{A}) \mathrm{d}A_1 \cdots \mathrm{d}A_n. \end{aligned} \tag{A5}$$

The forward Kolmogorov (Fökker-Planck) equation for the density is

$$\begin{aligned} p_T = \frac{1}{2}\varepsilon^2 \Delta^2 p_{SS} &+ \varepsilon^2 \sum_{i,j} \lambda_i \rho_{ij} \nu_j \left(A_i A_j p\right)_{SA_j} \\ &+ \frac{1}{2}\varepsilon^2 \sum_{i,j} r_{ij} \nu_i \nu_j \left(A_i A_j p\right)_{A_i A_j} \end{aligned} \tag{A6a}$$

for $T > t$, with the initial condition

$$p \to \delta(S - s)\delta^n(\mathbf{A} - \boldsymbol{\alpha}) \quad \text{as } T \to t. \tag{A6b}$$

The second and third terms are derivatives with respect to A_j, so integrating over all A_j yields zero:

$$\int \left(A_i A_j p\right)_{SA_j} \mathrm{d}A_j = 0, \quad \int \left(A_i A_j p\right)_{A_i A_j} \mathrm{d}A_j = 0. \tag{A7}$$

Consequently, if we integrate the forward Kolmogorov equation over all the A's, we obtain

$$Q_T^{(0)} = \tfrac{1}{2}\varepsilon^2 Q_{SS}^{(2)} \quad \text{for } T > t, \tag{A8a}$$

$$Q^{(0)}(T, S) \to \delta(S - s) \quad \text{as } T \to t. \tag{A8b}$$

The marginal density

$$Q^{(0)}(t, s, \boldsymbol{\alpha}; T, S) = \int \cdots \int p(t, s, \boldsymbol{\alpha}; T, S, \mathbf{A})\, \mathrm{d}A_1 \cdots \mathrm{d}A_n \tag{A9}$$

represents the probability that the value of $\tilde{S}(T)$ is S, regardless of values of the $\tilde{A}_j(T)$. Equation (A8a) is essentially a conservation equation for this marginal density. In the next section we use asymptotic methods to analyze the *backwards* Kolmogorov equation for the moments $Q^{(k)}$. This analysis will establish a 'constitutive' relationship that determines $Q^{(2)}$ in terms of $Q^{(0)}$,

$$Q^{(2)} = \Delta^2 \left(1 + 2\varepsilon\beta z + \varepsilon^2\gamma z^2 + \cdots\right) e^{\varepsilon^2\Gamma(T-t)} Q^{(0)}, \tag{A10}$$

where the coefficients are given by

$$z = \frac{S-s}{\varepsilon\Delta}, \tag{A11a}$$

$$\Delta^2 = \sum_{i,j} c_{ij}\lambda_i\lambda_j\alpha_i\alpha_j, \tag{A11b}$$

$$\beta = \frac{1}{\Delta^2}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\Delta_{\alpha_j}, \tag{A11c}$$

$$\gamma = \frac{1}{\Delta^2}\sum_{i,j} r_{ij}\nu_i\nu_j\alpha_i\alpha_j\Delta_{\alpha_i}\Delta_{\alpha_j} + \kappa, \tag{A11d}$$

$$\Gamma = \frac{1}{\Delta}\sum_{i,j} r_{ij}\nu_i\nu_j\alpha_i\alpha_j\Delta_{\alpha_i\alpha_j} - \kappa, \tag{A11e}$$

with

$$\kappa = \frac{1}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\beta_{\alpha_j}. \tag{A11f}$$

Although not exact, this relationship is accurate through $O(\varepsilon^2)$. Substituting this relationship into the conservation law then yields effective forward equation

$$Q_T^{(0)} = \tfrac{1}{2}\varepsilon^2\Delta^2 e^{\varepsilon^2\Gamma(T-t)}\left[\left(1 + 2\varepsilon\beta z + \varepsilon^2\gamma z^2\right)Q^{(0)}\right]_{SS} \quad \text{for } T > t. \tag{A12}$$

With the constitutive law in hand, we can now set $t = 0$ without loss of generality, and abbreviate

$$Q^{(0)}(T,S) \equiv Q^{(0)}(0,s,\boldsymbol{\alpha};T,S). \tag{A13}$$

The effective forward equation then becomes

$$Q_T^{(0)} = \tfrac{1}{2}\varepsilon^2\Delta^2 e^{\varepsilon^2\Gamma T}\left[\left(1 + 2\varepsilon\beta z + \varepsilon^2\gamma z^2\right)Q^{(0)}\right]_{SS} \quad \text{for } T > 0, \tag{A14a}$$

$$Q^{(0)}(T,S) \to \delta(S-s) \quad \text{as } T \to 0. \tag{A14b}$$

Even though the effective forward equation is not exact, *option prices obtained by numerically solving the effective forward equation are exactly arbitrage free*, since the maximum principle ensures that the probability density $Q^{(0)}$ is strictly positive for $T > 0$.

After establishing the effective forward equation, we shall find that there is a normal SABR model which has the same marginal density to within $O(\varepsilon^2)$. Consequently, European option prices obtained from this SABR model are also accurate to within $O(\varepsilon^2)$, the same accuracy as the original SABR equations.

A.2. The backwards equation

To derive the constitutive law, let

$$\tau = T - t. \tag{A15}$$

The backwards Kolmogorov equations for the moments $Q^{(k)}$ are

$$Q_\tau^{(k)} = \tfrac{1}{2}\varepsilon^2\Delta^2 Q_{ss}^{(k)} + \varepsilon^2\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j Q_{s\alpha_j}^{(k)} + \tfrac{1}{2}\varepsilon^2\sum_{i,j} r_{ij}\nu_i\nu_j\alpha_i\alpha_j Q_{\alpha_i\alpha_j}^{(k)} \tag{A16a}$$

for $\tau > 0$, with

$$Q^{(k)} \to \Delta^k(\boldsymbol{\alpha})\,\delta(S-s) \quad \text{as } \tau \to 0. \tag{A16b}$$

We proceed by repeatedly changing variables until the relationship between $Q^{(2)}$ and $Q^{(0)}$ is obvious. Let

$$z = \frac{S-s}{\varepsilon\Delta(\boldsymbol{\alpha})}. \tag{A17}$$

Changing variables from $s, \boldsymbol{\alpha}$ to $z, \boldsymbol{\alpha}$ requires

$$\frac{\partial}{\partial s} \to -\frac{1}{\varepsilon\Delta}\frac{\partial}{\partial z}, \quad \frac{\partial^2}{\partial s^2} \to \frac{1}{\varepsilon^2\Delta^2}\frac{\partial^2}{\partial z^2}, \tag{A18a}$$

$$\frac{\partial}{\partial\alpha_i} \to \frac{\partial}{\partial\alpha_i} - \frac{\Delta_{\alpha_i}}{\Delta}z\frac{\partial}{\partial z}, \tag{A18b}$$

$$\frac{\partial^2}{\partial s\partial\alpha_i} \to \frac{1}{\varepsilon\Delta}\left\{-\frac{\partial^2}{\partial z\partial\alpha_i} + \frac{\Delta_{a_i}}{\Delta}\left(z\frac{\partial^2}{\partial z^2} + \frac{\partial}{\partial z}\right)\right\}, \tag{A18c}$$

$$\frac{\partial^2}{\partial\alpha_i\partial\alpha_j} \to \frac{\partial^2}{\partial\alpha_i\partial\alpha_j} - \frac{\Delta_{\alpha_i}}{\Delta}z\frac{\partial^2}{\partial z\partial\alpha_j} - \frac{\Delta_{\alpha_j}}{\Delta}z\frac{\partial^2}{\partial z\partial\alpha_i} + \frac{\Delta_{\alpha_i}\Delta_{\alpha_j}}{\Delta^2}z^2\frac{\partial^2}{\partial z^2} + \frac{2\Delta_{\alpha_i}\Delta_{\alpha_j} - \Delta\Delta_{\alpha_i\alpha_j}}{\Delta^2}z\frac{\partial}{\partial z}, \tag{A18d}$$

and

$$\delta(S-s) \to \delta(\varepsilon\Delta(\boldsymbol{\alpha})\,z) = \frac{1}{\varepsilon\Delta(\boldsymbol{\alpha})}\delta(z). \tag{A18e}$$

In the new variables, the backwards equation becomes

$$Q_\tau^{(k)} = \tfrac{1}{2}\left(1 + 2\varepsilon\beta z + \varepsilon^2 g z^2\right)Q_{zz}^{(k)} + \varepsilon\beta Q_z^{(k)} + \varepsilon^2\left(g - \tfrac{1}{2}\phi\right)zQ_z^{(k)} - \frac{\varepsilon}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j Q_{z\alpha_j}^{(k)} + \frac{1}{2}\varepsilon^2\sum_{i,j} r_{ij}\nu_i\nu_j\alpha_i\alpha_j\left\{Q_{\alpha_i\alpha_j}^{(k)} - 2\frac{\Delta_{\alpha_i}}{\Delta}zQ_{z\alpha_j}^{(k)}\right\}, \tag{A19a}$$

with

$$Q^{(k)} \to \frac{1}{\varepsilon}\Delta^{k-1}(\boldsymbol{\alpha})\,\delta(z) \quad \text{as } \tau \to 0. \tag{A19b}$$

Here,

$$\beta(\boldsymbol{\alpha}) = \frac{1}{\Delta^2}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\Delta_{\alpha_j}, \tag{A20a}$$

$$g(\boldsymbol{\alpha}) = \frac{1}{\Delta^2}\sum_{i,j} r_{ij}\nu_i\nu_j\alpha_i\alpha_j\Delta_{\alpha_i}\Delta_{\alpha_j}, \tag{A20b}$$

$$\phi(\boldsymbol{\alpha}) = \frac{1}{\Delta}\sum_{i,j} r_{ij}\nu_i\nu_j\alpha_i\alpha_j\Delta_{\alpha_i\alpha_j}. \tag{A20c}$$

Next, define $\theta^{(k)}$ via

$$Q^{(k)} = \frac{1}{\varepsilon}\Delta^{k-1}(\boldsymbol{\alpha})\,\theta^{(k)}. \tag{A21}$$

Then

$$Q_{z\alpha_i}^{(k)} \to \theta_{z\alpha_i}^{(k)} + (k-1)\frac{\Delta_{\alpha_i}}{\Delta}\theta_z^{(k)}, \tag{A22a}$$

$$Q_{\alpha_i\alpha_j}^{(k)} \to \theta_{\alpha_i\alpha_j}^{(k)} + \frac{k-1}{\Delta}\left(\Delta_{\alpha_i}\theta_{\alpha_j}^{(k)} + \Delta_{\alpha_j}\theta_{\alpha_i}^{(k)} + \Delta_{\alpha_i\alpha_j}\theta^{(k)}\right) + \frac{(k-1)(k-2)}{\Delta^2}\Delta_{\alpha_i}\Delta_{\alpha_j}\theta^{(k)}. \tag{A22b}$$

The backwards equation becomes

$$\begin{aligned}\theta_\tau^{(k)} &= \frac{1}{2}\left(1+2\varepsilon\beta z+\varepsilon^2 g z^2\right)\theta_{zz}^{(k)} \\ &\quad - \varepsilon\,(k-2)\,\beta\theta_z^{(k)} - \varepsilon^2\,(k-2)\,gz\theta_z^{(k)} \\ &\quad + \tfrac{1}{2}\varepsilon^2\,(k-1)\,(k-2)\,g\theta^{(k)} - \tfrac{1}{2}\varepsilon^2\phi\left\{z\theta_z^{(k)} - (k-1)\,\theta^{(k)}\right\} \\ &\quad - \frac{\varepsilon}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\theta_{z\alpha_j}^{(k)} \\ &\quad + \tfrac{1}{2}\varepsilon^2\sum_{i,j} r_{ij}\nu_i\nu_j\alpha_i\alpha_j \\ &\quad \times\left\{\theta_{\alpha_i\alpha_j}^{(k)} + 2\,(k-1)\,\frac{\Delta_{\alpha_i}}{\Delta}\theta_{\alpha_j}^{(k)} - 2\frac{\Delta_{\alpha_i}}{\Delta}z\theta_{z\alpha_j}^{(k)}\right\},\end{aligned} \tag{A23a}$$

with

$$\theta^{(k)} \to \delta(z) \quad \text{as } \tau\to 0. \tag{A23b}$$

This equation and initial condition are independent of $\boldsymbol{\alpha}$ to leading order. Therefore, $\theta^{(k)}$ is indepedent of $\boldsymbol{\alpha}$ to leading order. This means that $\varepsilon^2\theta_{\alpha_i\alpha_j}^{(k)}$, $\varepsilon^2\theta_{\alpha_j}^{(k)}$, and $\varepsilon^2\theta_{z\alpha_j}^{(k)}$ are all no larger than $O(\varepsilon^3)$. As we are only working through $O(\varepsilon^2)$, we neglect these terms, obtaining

$$\begin{aligned}\theta_\tau^{(k)} &= \tfrac{1}{2}\left(1+2\varepsilon\beta z+\varepsilon^2 g z^2\right)\theta_{zz}^{(k)} \\ &\quad - \varepsilon\,(k-2)\,\beta\theta_z^{(k)} - \varepsilon^2\,(k-2)\,gz\theta_z^{(k)} \\ &\quad + \tfrac{1}{2}\varepsilon^2\,(k-1)\,(k-2)\,g\theta^{(k)} - \tfrac{1}{2}\varepsilon^2\phi\left\{z\theta_z^{(k)} - (k-1)\,\theta^{(k)}\right\} \\ &\quad - \frac{\varepsilon}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\theta_{z\alpha_j}^{(k)},\end{aligned} \tag{A24a}$$

$$\theta^{(k)} \to \delta(z) \quad \text{as } \tau\to 0, \tag{A24b}$$

through $O(\varepsilon^2)$.

For the special cases of $k=0$ and $k=2$, this yields

$$\begin{aligned}\theta_\tau^{(0)} &= \tfrac{1}{2}\left[\left(1+2\varepsilon\beta z+\varepsilon^2 g z^2\right)\theta^{(0)}\right]_{zz} - \tfrac{1}{2}\varepsilon^2\phi\left\{z\theta_z^{(0)}+\theta^{(0)}\right\} \\ &\quad - \frac{\varepsilon}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\theta_{z\alpha_j}^{(0)},\end{aligned} \tag{A25a}$$

$$\begin{aligned}\theta_\tau^{(2)} &= \tfrac{1}{2}\left(1+2\varepsilon\beta z+\varepsilon^2 g z^2\right)\theta_{zz}^{(2)} - \tfrac{1}{2}\varepsilon^2\phi\left\{z\theta_z^{(2)}-\theta^{(2)}\right\} \\ &\quad - \frac{\varepsilon}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\theta_{z\alpha_j}^{(2)},\end{aligned} \tag{A25b}$$

$$\theta^{(0)}\to\delta(z), \quad \theta^{(2)}\to\delta(z) \quad \text{as } \tau\to 0. \tag{A25c}$$

The equations for $\theta^{(2)}$ and $\theta^{(0)}$ are very similar; we now transform $\theta^{(0)}$ until they are identical, at least through $O(\varepsilon^2)$. Define

$$U = \left(1+2\varepsilon\beta z+\varepsilon^2 g z^2\right)\theta^{(0)} \tag{A26}$$

Then Equation (A25a) becomes

$$\begin{aligned}U_\tau &= \tfrac{1}{2}\left(1+2\varepsilon\beta z+\varepsilon^2 g z^2\right)U_{zz} - \tfrac{1}{2}\varepsilon^2\phi\left\{zU_z - U\right\} \\ &\quad - \frac{\varepsilon}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j U_{z\alpha_j} \\ &\quad + 2\varepsilon^2\kappa\,(zU)_z - \varepsilon^2\phi U,\end{aligned} \tag{A27a}$$

$$U\to\delta(z) \quad \text{as } \tau\to 0, \tag{A27b}$$

through $O(\varepsilon^2)$, where

$$\kappa = \frac{1}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\beta_{\alpha_j}. \tag{A28}$$

Finally, we set

$$U = He^{-\varepsilon^2\kappa z^2-\varepsilon^2(\phi-\kappa)\tau}. \tag{A29}$$

In terms of H, we obtain

$$\begin{aligned}H_\tau &= \frac{1}{2}\left(1+2\varepsilon\beta z+\varepsilon^2 g z^2\right)H_{zz} - \tfrac{1}{2}\varepsilon^2\phi\left\{zH_z - H\right\} \\ &\quad - \frac{\varepsilon}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j H_{z\alpha_j},\end{aligned} \tag{A30a}$$

$$H\to\delta(z) \quad \text{as } \tau\to 0, \tag{A30b}$$

through $O(\varepsilon^2)$. This is identical to equations (A25b) and (A25c) for $\theta^{(2)}$. Therefore, H is identical to $\theta^{(2)}$, at least through $O(\varepsilon^2)$. Tracking back through the transformations, this means that

$$\begin{aligned}Q^{(2)} &\equiv \Delta^2(\boldsymbol{\alpha})\,e^{\varepsilon^2\kappa z^2+\varepsilon^2(\phi-\kappa)\tau}\left(1+2\varepsilon\beta z+\varepsilon^2 g z^2\right)Q^{(0)} \\ &= \Delta^2(\boldsymbol{\alpha})\,e^{\varepsilon^2\Gamma\tau}\left(1+2\varepsilon\beta z+\varepsilon^2\gamma z^2\right)Q^{(0)},\end{aligned} \tag{A31a}$$

through $O(\varepsilon^2)$, with

$$\gamma = g+\kappa, \quad \Gamma = \phi-\kappa. \tag{A31b}$$

Substituting this into our conservation equation (A8a), we obtain

$$Q_\tau^{(0)} = \tfrac{1}{2}\varepsilon^2\Delta^2 e^{\varepsilon^2\Gamma\tau}\left[\left(1+2\varepsilon\beta z+\varepsilon^2\gamma z^2\right)Q^{(0)}\right]_{SS} \quad \text{for } \tau>0.$$

Having obtained the effective forward equation, we set $t=0$ without loss of generality. Then the marginal density $Q^{(0)}(T,S)\equiv Q^{(0)}(0,s,\boldsymbol{\alpha};T,S)$ satisfies

$$Q_T^{(0)} = \tfrac{1}{2}\varepsilon^2\Delta^2 e^{\varepsilon^2\Gamma T}\left[\left(1+2\varepsilon\beta z+\varepsilon^2\gamma z^2\right)Q^{(0)}\right]_{SS} \quad \text{for } T>0, \tag{A32a}$$

$$Q^{(0)}(T,S)\to\delta(S-s) \quad \text{as } T\to 0. \tag{A32b}$$

Here,

$$z = \frac{S-s}{\varepsilon\Delta}, \tag{A33a}$$

where

$$\Delta^2(\boldsymbol{\alpha}) = \sum_{i,j} c_{ij}\lambda_i\alpha_i\lambda_j\alpha_j, \tag{A33b}$$

$$\begin{aligned}\beta(\boldsymbol{\alpha}) &= \frac{1}{\Delta^2}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\Delta_{\alpha_j} \\ &= \frac{1}{\Delta^3}\sum_{i,j,k}\rho_{ij}\nu_j c_{jk}\lambda_i\alpha_i\lambda_j\alpha_j\lambda_k\alpha_k,\end{aligned} \tag{A33c}$$

$$\begin{aligned}\gamma(\boldsymbol{\alpha}) &= \frac{1}{\Delta^2}\sum_{i,j} r_{ij}\nu_i\nu_j\alpha_i\alpha_j\Delta_{\alpha_i}\Delta_{\alpha_j} + \kappa(\boldsymbol{\alpha}) \\ &= \frac{1}{\Delta^4}\sum_{i,j,k,m} r_{ij}\nu_i\nu_j c_{ik}c_{jm}\lambda_i\alpha_i\lambda_j\alpha_j\lambda_k\alpha_k\lambda_m\alpha_m + \kappa(\boldsymbol{\alpha}),\end{aligned} \tag{A33d}$$

$$\begin{aligned}\Gamma(\boldsymbol{\alpha}) &= \frac{1}{\Delta}\sum_{i,j} r_{ij}\nu_i\nu_j\alpha_i\alpha_j\Delta_{\alpha_i\alpha_j} - \kappa(\boldsymbol{\alpha}) \\ &= \frac{1}{\Delta^4}\sum_{i,j,k,m} r_{ij}\nu_i\nu_j\left(c_{ij}c_{km}-c_{ik}c_{jm}\right) \\ &\quad \times\lambda_i\alpha_i\lambda_j\alpha_j\lambda_k\alpha_k\lambda_m\alpha_m - \kappa(\boldsymbol{\alpha}),\end{aligned} \tag{A33e}$$

with

$$\kappa(\boldsymbol{\alpha}) = \frac{1}{\Delta}\sum_{i,j}\lambda_i\rho_{ij}\nu_j\alpha_i\alpha_j\beta_{\alpha_j}$$
$$= \frac{1}{\Delta^4}\sum_{i,j,k,m}\rho_{ij}\nu_j\left(\rho_{ki}\nu_i + \rho_{kj}\nu_j + \rho_{km}\nu_m\right)$$
$$\times c_{jm}\lambda_i\alpha_i\lambda_j\alpha_j\lambda_k\alpha_k\lambda_m\alpha_m - 3\beta^2(\boldsymbol{\alpha}). \tag{A33f}$$

A.3. *The effective SABR model*

There is a normal SABR model which has the same marginal density $Q^{(0)}$ to within $O(\varepsilon^2)$. Consequently, European option prices obtained from this SABR model are also accurate to within $O(\varepsilon^2)$, the same accuracy as the original SABR equations. The caveat is that for each expiry date t_{ex}, a slightly different value of the SABR vol α is needed to match the marginal density.

To show this, recall that in Hagan *et al.* (2013) it is shown that if the forward price of an asset is driven by the normal SABR model,

$$\mathrm{d}\tilde{S} = \varepsilon\tilde{A}\mathrm{d}W, \tag{A34a}$$
$$\mathrm{d}\tilde{A} = \varepsilon\nu\tilde{A}\mathrm{d}Z, \tag{A34b}$$
$$\mathrm{d}Z\,\mathrm{d}W = \rho\mathrm{d}T, \tag{A34c}$$

then the marginal probability density

$$Q^{SABR}(T,S) = \mathrm{E}\left\{\delta\left(\tilde{S}(T) - S\right)\middle|\,\tilde{S}(0) = s, \tilde{A}(0) = \alpha\right\}$$
$$= \int p(0,s,\alpha;T,S,A)\,\mathrm{d}A \tag{A35}$$

satisfies the effective forward equation

$$Q_T^{SABR} = \tfrac{1}{2}\varepsilon^2\alpha^2\left[\left(1 + 2\varepsilon\rho\nu z + \varepsilon^2\nu^2z^2\right)Q^{SABR}\right]_{SS} \quad \text{for } T > 0, \tag{A36a}$$
$$Q^{SABR}(T,S) \to \delta(S - s) \quad \text{as } T \to 0, \tag{A36b}$$

through $O(\varepsilon^2)$. To bring this in line with A32a, we define the new time variable

$$T_{new} = \frac{1 - e^{-\varepsilon^2\Gamma T}}{\varepsilon^2\Gamma C}, \tag{A37a}$$

where for any given expiry date t_{ex}, C is chosen as

$$C = \frac{1 - e^{-\varepsilon^2\Gamma t_{ex}}}{\varepsilon^2\Gamma t_{ex}} = e^{-\frac{1}{2}\varepsilon^2\Gamma t_{ex}}\left\{1 - \tfrac{1}{24}\varepsilon^4\Gamma^2t_{ex}^2 + \cdots\right\}, \tag{A37b}$$

so that $T_{new}(t_{ex}) = t_{ex}$. In the new time variable, the SABR model's effective forward equation becomes

$$Q_T^{SABR} = \tfrac{1}{2}\varepsilon^2C\alpha^2e^{\varepsilon^2\Gamma T}\left[\left(1 + 2\varepsilon\rho\nu z + \varepsilon^2\nu^2z^2\right)Q^{SABR}\right]_{SS}$$
$$\text{for } T > 0, \tag{A38a}$$
$$Q^{SABR}(T,S) \to \delta(S - s) \quad \text{as } T \to 0, \tag{A38b}$$

where we have dropped the 'new' subscript for brevity. If we select

$$\alpha_{eff} = C^{-1/2}\Delta(\boldsymbol{\alpha}) \equiv \Delta(\boldsymbol{\alpha})\,e^{\frac{1}{4}\varepsilon^2\Gamma t_{ex}}, \tag{A39a}$$
$$\rho_{eff} = \beta/\sqrt{\gamma}, \tag{A39b}$$
$$\nu_{eff} = \sqrt{\gamma}, \tag{A39c}$$

then Equations (A38a) and (A38b) for Q^{SABR} exactly match Equations (A32a) and (A32b) for $Q^{(0)}$, so the solutions of these equations must also match exactly. Consequently, at t_{ex} the SABR model's marginal density $Q^{SABR}(t_{ex},S)$ must match $Q^{(0)}(t_{ex},S)$ through $O(\varepsilon^2)$. Thus, all European options with exercise date t_{ex} are correctly valued by the SABR model through $O(\varepsilon^2)$, where the SABR parameters $\alpha_{eff} = \tilde{A}(0)$, ρ_{eff}, ν_{eff} are given by Equations (A39a)–(A39c) with (A33b)–(A33f).

One can evaluate European options, either standard options (Hagan *et al.* 2016) or quadratic options (Hagan *et al.* 2019), by solving the effective forward equation numerically (Hagan *et al.* 2013), or by using the explicit implied volatility formulas for the SABR model contained in Hagan *et al.* (2016) and Hagan *et al.* (2019).

A.4. *Special cases*

The disadvantage of the above results is that the formulas rely on too many parameters. So we analyze two special cases which have a more parsimonious parameter set. These special cases only require knowing the SABR models for the individual components $\tilde{R}_j$ and knowing the asset-to-asset correlation matrix, c_{ij}, which is observable. The two cases represent opposites: in one case the volatilities are correlated as highly as possible, and in the other they are as independent as possible. Because of its theoretical importance, we also include the case where the assets are statistically identical.

Before continuing, we note that we can simplify these cases further by using Markowitz-type model to obtain c_{ij}. In such models, each asset $\tilde{R}_j$ is driven by the market as a whole and by its own idiosyncratic noise. So,

$$\mathrm{d}W_j = \eta_j\mathrm{d}W_{mkt} + \sqrt{1 - \eta_j^2}\,\mathrm{d}W_j^{\perp}, \tag{A40a}$$

where

$$\mathrm{d}W_j^{\perp}\mathrm{d}W_{mkt} = \mathrm{d}W_i^{\perp}\mathrm{d}W_j^{\perp} = 0 \quad \text{for all } i,j. \tag{A40b}$$

For Markowitz models, the $\tilde{R}$-to-$\tilde{R}$ correlation matrix is

$$c_{ij} = \begin{cases} 1 & \text{for } i = j \\ \eta_i\eta_j & \text{for} i \neq j \end{cases}. \tag{A41}$$

A.4.1. Single stochastic vol.

In most markets volatilities are highly correlated; the entire market tends to be volatile or quiet in synchrony. If we neglect the idiosyncrasies of the individual vols, we can model all the stochastic volatilities with a single process

$$\mathrm{d}Z_1 = \mathrm{d}Z_2 = \cdots = dZ_n \equiv \mathrm{d}Z. \tag{A42}$$

This implies that

$$\rho_{ij} = \rho_{ii} \quad \text{for all } j, \tag{A43a}$$
$$r_{ij} = 1 \quad \text{for all } i,j. \tag{A43b}$$

For this case the effective normal ($\beta_{eff} = 0$) SABR model simplifies to

$$\alpha_{eff} = \Delta e^{\frac{1}{4}\varepsilon^2\Gamma t_{ex}}, \tag{A44a}$$
$$\rho_{eff} = \bar{\rho}\bar{\nu}/\sqrt{\bar{\nu}^2 + \kappa}, \tag{A44b}$$
$$\nu_{eff} = \sqrt{\bar{\nu}^2 + \kappa}, \tag{A44c}$$

where

$$\Delta^2(\boldsymbol{\alpha}) = \sum_{i,j}c_{ij}\lambda_i\alpha_i\lambda_j\alpha_j, \tag{A45a}$$
$$\bar{\rho}(\boldsymbol{\alpha}) = \frac{1}{\Delta}\sum_i\rho_i\lambda_i\alpha_i, \tag{A45b}$$
$$\bar{\nu}(\boldsymbol{\alpha}) = \frac{1}{\Delta^2}\sum_{i,j}c_{ij}\lambda_i\alpha_i\lambda_j\alpha_j\nu_j, \tag{A45c}$$

and

$$\Gamma(\boldsymbol{\alpha}) = \frac{1}{\Delta^2}\sum_{i,j} c_{ij}(\nu_i - \bar{\nu})(\nu_j - \bar{\nu})\lambda_i\alpha_i\lambda_j\alpha_j - \kappa(\boldsymbol{\alpha}), \quad \text{(A45d)}$$

$$\kappa(\boldsymbol{\alpha}) = \frac{\bar{\rho}\bar{\nu}}{\Delta}\sum_i \lambda_i\alpha_i\rho_i(\nu_i - \bar{\nu}) + \tfrac{1}{2}\frac{\bar{\rho}^2}{\Delta^2}\sum_{i,j} c_{ij}\lambda_i\alpha_i\lambda_j\alpha_j(\nu_i + \nu_j - 2\bar{\nu})^2. \quad \text{(A45e)}$$

Note that $\bar{\rho}$ and $\bar{\nu}$ represent the volatility weighted correlation and vol-of-vol, while Γ and κ represent the dispersion of vols across the different assets. It is not uncommon for the vols ν_i to be very similar to each other, so that Γ and κ are much smaller than $\bar{\nu}^2$. In this case $\gamma \simeq g$ and $\Gamma \simeq 0$, so the effective SABR coefficients are roughly

$$\alpha_{eff} = \Delta e^{\frac{1}{4}\varepsilon^2\Gamma t_{ex}} \simeq \Delta(\boldsymbol{\alpha}), \quad \text{(A46a)}$$

$$\rho_{eff} = \bar{\rho}\bar{\nu}/\sqrt{\bar{\nu}^2 + \kappa} \simeq \bar{\rho}, \quad \text{(A46b)}$$

$$\nu_{eff} = \sqrt{\bar{\nu}^2 + \kappa} \simeq \bar{\nu}. \quad \text{(A46c)}$$

A.4.2. Independent stochastic volatilities. The opposite extreme is one in which the volatilities are as independent as possible. To match each asset's volatility smile, we need to allow the volatilities to be correlated with their own assets, $\mathrm{d}W_j\mathrm{d}Z_j = \rho_{jj}\mathrm{d}T$. If the asset prices are correlated with each other, $\mathrm{d}W_i\mathrm{d}W_j = c_{ij}\mathrm{d}T$, it is unreasonable to expect the stochastic processes are uncorrelated with each other. Instead we assume that

$$\mathrm{d}Z_j = \rho_{jj}\mathrm{d}W_j + \sqrt{1-\rho_{jj}^2}\,\mathrm{d}Z_j^{\perp} \quad \text{(A47)}$$

where the Brownian motions $dZ_j^{\perp}$ are independent of all other stochastic processes. This implies that

$$\rho_{ij} = c_{ij}\rho_{jj} \quad \text{for all } i,j, \quad \text{(A48a)}$$

$$r_{ij} = c_{ij}\rho_{ii}\rho_{jj} \quad \text{for all } i,j. \quad \text{(A48b)}$$

The coefficients for the effective forward equation and the parameters for the effective SABR model are then found by substituting (A48a) and (A48b) into (A39a), (A39c), (A33a)–(A33f). There is no simplification of these formulas, so we omit writing these results explicitly.

A.4.3. Statistically identical assets. The last special case is one where all the assets in the basket are modeled as being statistically identical. Then,

$$\mathrm{d}W_i = \sqrt{c}\,\mathrm{d}W_{mkt} + \sqrt{1-c}\,\mathrm{d}W_i^{\perp}, \quad \text{(A49a)}$$

$$\mathrm{d}Z_i = \sqrt{r}\,\mathrm{d}Z_{mkt} + \sqrt{1-r}\,\mathrm{d}Z_i^{\perp}, \quad \text{(A49b)}$$

so

$$\mathrm{d}W_i\mathrm{d}Z_i = \sqrt{cr}\,\rho_{mkt} + \sqrt{(1-c)(1-r)}\,\rho^{\perp}, \quad \text{(A50a)}$$

$$\mathrm{d}W_i\mathrm{d}Z_j = \sqrt{cr}\,\rho_{mkt}. \quad \text{(A50b)}$$

Without loss of generality, we set $\lambda_i \equiv 1$, so

$$\alpha_i \equiv \alpha, \quad \nu_j = \nu \quad \text{for all } i, \quad \text{(A51a)}$$

$$c_{ij} = \begin{cases} 1 & \text{for } i = j \\ c & \text{for } i \neq j \end{cases}, \quad r_{ij} = \begin{cases} 1 & \text{for } i = j \\ r & \text{for } i \neq j \end{cases}, \quad \text{(A51b)}$$

$$\rho_{ij} = \begin{cases} \rho_{\shortparallel} & \text{for } i = j \\ \rho_{\perp} & \text{for } i \neq j \end{cases} \quad \text{(A51c)}$$

Substituting this into (A33a)–(A33f) eventually yields,

$$\kappa = 0, \quad \text{(A52)}$$

so the coefficients reduce to

$$\Delta^2 = \alpha^2 n(1 - c + nc), \quad \text{(A53a)}$$

$$\beta = \frac{\rho_{\shortparallel} - \rho_{\perp} + n\rho_{\perp}}{n^{1/2}(1 - c + nc)^{1/2}}\nu, \quad \text{(A53b)}$$

$$\gamma = \frac{1 - r + nr}{n}\nu^2, \quad \text{(A53c)}$$

$$\Gamma = \frac{(n-1)(1-r)(1-c)}{n(1-c+nc)}\nu^2, \quad \text{(A53d)}$$

and the SABR coefficients are

$$\alpha_{eff} = \alpha n^{1/2}(1 - c + nc)^{1/2} e^{\frac{1}{4}\varepsilon^2\Gamma t_{ex}}, \quad \text{(A54a)}$$

$$\rho_{eff} = \frac{\rho_{\shortparallel} - \rho_{\perp} + n\rho_{\perp}}{(1 - c + nc)^{1/2}(1 - r + nr)^{1/2}}, \quad \text{(A54b)}$$

$$\nu_{eff} = \sqrt{\frac{1 - r + nr}{n}}\,\nu. \quad \text{(A54c)}$$

If, in addition, we assume that there is one stochastic vol, so $dZ_i \equiv dZ$ for all i, then the formula for γ simplifies and Γ is zero,

$$\gamma = \frac{1 - r + nr}{n}\nu^2, \quad \text{(A55a)}$$

$$\Gamma = 0. \quad \text{(A55b)}$$

Consequently, the effective coefficients for the normal SABR model are

$$\alpha_{eff} = an^{1/2}(1 - c + nc)^{1/2}, \quad \text{(A56a)}$$

$$\rho_{eff} = \frac{\rho_{\shortparallel} - \rho_{\perp} + n\rho_{\perp}}{(1 - c + nc)^{1/2}}, \quad \text{(A56b)}$$

$$\nu_{eff} = \nu. \quad \text{(A56c)}$$

Appendix 2. The effective SABR model for spreads

We now specialize to $n = 2$ with $\lambda_1 = 1$ and $\lambda_2 = -1$, so that

$$\tilde{S} = \tilde{R}_1 - \tilde{R}_2 \quad \text{(A57)}$$

is the spread between the swap rates. With only two components, we set

$$\mathrm{d}W_1\mathrm{d}W_2 = c\,\mathrm{d}T, \quad \mathrm{d}Z_1\mathrm{d}Z_2 = r\,\mathrm{d}T, \quad \text{(A58a)}$$

and only ρ_{ij} is a full matrix:

$$\mathrm{d}W_i\mathrm{d}Z_j = \rho_{ij}\mathrm{d}T \quad \text{for } i,j = 1,2. \quad \text{(A58b)}$$

In this case the SABR model parameters (A39a)–(A39c) with (A33b)–(A33f) become

$$\alpha_{eff} = \Delta(\boldsymbol{\alpha})\, e^{\frac{1}{4}\varepsilon^2\Gamma t_{ex}}, \quad \text{(A59a)}$$

$$\rho^{eff} = \beta(\boldsymbol{\alpha})/\sqrt{\gamma(\boldsymbol{\alpha})}, \quad \text{(A59b)}$$

$$\nu^{eff} = \sqrt{\gamma(\boldsymbol{\alpha})}, \quad \text{(A59c)}$$

with

$$\Delta^2(\boldsymbol{\alpha}) = \alpha_1^2 - 2c\alpha_1\alpha_2 + \alpha_2^2, \quad \text{(A60a)}$$

$$\beta(\boldsymbol{\alpha}) = \rho_1^{eff}\nu_1^{eff} + \rho_2^{eff}\nu_2^{eff}, \quad \text{(A60b)}$$

$$\gamma(\boldsymbol{\alpha}) = \left(\nu_1^{eff}\right)^2 + 2r\nu_1^{eff}\nu_2^{eff} + \left(\nu_2^{eff}\right)^2 + \kappa(\boldsymbol{\alpha}), \quad \text{(A60c)}$$

$$\Gamma(\boldsymbol{\alpha}) = \left(1 - c^2\right)\frac{\alpha_1^2\alpha_2^2}{\Delta^4}\left(\nu_1^2 - 2r\nu_1\nu_2 + \nu_2^2\right) - \kappa(\boldsymbol{\alpha}),$$

$$\kappa(\boldsymbol{\alpha}) = \left\{3\left(1 - c^2\right)\frac{\alpha_1\alpha_2}{\Delta^2} - 2c\right\}\frac{\alpha_1\alpha_2}{\Delta^2}\nu_{mix}^2 \quad \text{(A60d)}$$

$$+ \frac{1}{\Delta}\left(\alpha_1\rho_{12}\nu_2^{eff} + \alpha_2\rho_{21}\nu_1^{eff}\right)\nu_{mix}, \quad \text{(A60e)}$$

where we have defined

$$\rho_1^{eff} \equiv \frac{\alpha_1\rho_{11} - \alpha_2\rho_{21}}{\Delta}, \quad \rho_2^{eff} \equiv \frac{\alpha_1\rho_{12} - \alpha_2\rho_{22}}{\Delta}, \tag{A61a}$$

$$\nu_1^{eff} \equiv \frac{(\alpha_1 - c\alpha_2)\,\alpha_1}{\Delta^2}\nu_1, \quad \nu_2^{eff} \equiv \frac{(\alpha_2 - c\alpha_1)\,\alpha_2}{\Delta^2}\nu_2, \tag{A61b}$$

$$\nu_{mix} = \frac{(\alpha_1\rho_{11} - \alpha_2\rho_{21})\,\nu_1 - (\alpha_1\rho_{12} - \alpha_2\rho_{22})\,\nu_2}{\Delta}. \tag{A61c}$$

A.5. Single stochastic vol for the spread

If both volatilities are driven by the same Brownian motion, $dZ_1 = dZ_2 \equiv dZ$, then

$$\rho_{12} = \rho_{11}, \quad \rho_{21} = \rho_{22}, \quad r \equiv 1. \tag{A62}$$

In this case the SABR parameters are

$$\alpha_{eff} = \Delta e^{\frac{1}{4}\varepsilon^2\Gamma t_{ex}}, \tag{A63a}$$

$$\rho_{eff} = \bar{\rho}\bar{\nu}/\sqrt{\bar{\nu}^2 + \kappa}, \tag{A63b}$$

$$\nu_{eff} = \sqrt{\bar{\nu}^2 + \kappa}, \tag{A63c}$$

with

$$\Delta^2 = \alpha_1^2 - 2c\alpha_1\alpha_2 + \alpha_2^2, \tag{A64a}$$

$$\bar{\rho} = \frac{\alpha_1\rho_{11} - \alpha_2\rho_{22}}{\Delta}, \tag{A64b}$$

$$\bar{\nu} = \frac{\alpha_1^2\nu_1 - c\alpha_1\alpha_1\,(\nu_1 + \nu_2) + \alpha_2^2\nu_2}{\Delta^2}, \tag{A64c}$$

and

$$\Gamma(\boldsymbol{\alpha}) = \left(1 - c^2\right)\frac{\alpha_1^2\alpha_2^2}{\Delta^4}\,(\nu_1 - \nu_2)^2 - \kappa(\boldsymbol{\alpha}), \tag{A65a}$$

$$\begin{aligned} \kappa(\boldsymbol{\alpha}) = {} & \left\{3\left(1 - c^2\right)\frac{\alpha_1^2\alpha_2^2}{\Delta^4} - c\frac{\alpha_1\alpha_2}{\Delta^2}\right\}\bar{\rho}^2\,(\nu_1 - \nu_2)^2 \\ & + \frac{\alpha_1\alpha_2}{\Delta^3}\,\{\alpha_1\,(\rho_{22} - c\rho_{11})\,\nu_1 \\ & + \alpha_2\,(\rho_{11} - c\rho_{22})\,\nu_2\}\,\bar{\rho}\,(\nu_1 - \nu_2). \end{aligned} \tag{A65b}$$

A.6. Independent stochastic volatilities

We can make the stochastic volatilities as independent as possible by assuming that

$$dZ_j = \rho_{jj}\,dW_j + \sqrt{1 - \rho_{jj}^2}\,dZ_j^{\perp} \tag{A66}$$

where the Brownian motions $dZ_j^{\perp}$ are independent of all other stochastic processes. This implies that

$$\rho_{12} = c\rho_{22}, \quad \rho_{21} = c\rho_{11}, \tag{A67a}$$

$$r_{11} = \rho_{11}^2, \quad r_{12} = c\rho_{11}\rho_{22}, \quad r_{22} = \rho_{22}^2. \tag{A67b}$$

The coefficients for the effective forward equation and the parameters for the basket SABR model are then found by substituting (A67a) and (A67b) into (A59a)–(A61c). There is no simplification of these formulas, so we omit writing these results explicitly.

Appendix 3. Shifters/ converters

In the current low rate environment, most trading desks use the shifted SABR model,

$$d\tilde{R} = \varepsilon\tilde{A}\left(\tilde{R} + s\right)^{\beta} dW + \text{drift terms}, \tag{A68a}$$

$$d\tilde{A} = \varepsilon\nu\tilde{A}dZ, \tag{A68b}$$

$$dW\,dZ = \rho\,dT, \tag{A68c}$$

to handle swaption and caplet smiles. As information is traded among different desks and counterparties, each with their own opinion of the best exponents β and offsets s to use, desks often need to convert SABR models with one value of β_1 and s_1 to SABR models with a different value of β_2 and s_2. A *shifter* or *converter* is the code that does this. This code first calculates the implied vols from the SABR model with β_1 and s_1 at a set of specific set of strikes. (A common set of strikes are strikes with deltas of 10%, 25%, 50%, 75%, 90%, where the deltas are determined by Bartlett's method Bartlett 2006). It then uses a weighted least squares to fit this smile to the SABR parameters with $\beta = \beta_2$ and shift s_2.

A key requirement of shifters is that converting a SABR model from β_1, s_1 to β_2, s_2, and then converting it back to β_1, s_1 should lead *exactly* to the original model. To meet this requirement, conversion is done in two steps. The first step is to use weighted least squares to fit the starting SABR model to a reference normal SABR model with $\beta = 0$. The second step is to find the SABR model with β_2, s_2 whose least squares fit to $\beta = 0$ matches this reference SABR model exactly. I.e. the second step is essentially the inverse of the first step. Since the second step is the inverse of the first step, it is usually done by first finding a trial SABR model by using weighted least squares to fit the SABR model with β_2, s_2 to the smile of the *reference* SABR model, and then iterating to find the SABR model with β_2, s_2 which converts exactly to the reference model.